State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on S^3

J. Loutey*

June 2026 (preprint)

Abstract

[INTERNAL THEOREM] We prove, unconditionally, that the Connes–van Suijlekom truncations of the round $S^3 = \mathrm{SU}(2)$ Camporesi–Higuchi spectral triple converge to the continuum spectral triple in van Suijlekom's state-space Gromov–Hausdorff distance [8]: the bound $d_{\mathrm{GH}} \leq \gamma_{n_{\max}}$, with $\gamma_{n_{\max}}$ the mass-concentration moment of the central spectral Fejér kernel on SU(2), and $\gamma_{n_{\max}} \to 0$ are proved. The displayed rate $\gamma_{n_{\max}} = (4/\pi + o(1)) \log n_{\max}/n_{\max}$ (dual-Coxeter normalisation of the SU(2) metric; equivalently $2/\pi$ on the unit round S^3; the constant scales with the metric, 2026-09-03 convention note) is **[INTERNAL THEOREM + MEASURED]**: derived, with its two logarithmic sub-coefficients pinned by doubling estimators (Appendix A). **[INTERNAL THEOREM]** The truncated state spaces are metrized by the left-translation Lipschitz seminorm, which equals the Dirac/metric Lipschitz constant on the continuum and whose kernel at every finite cutoff is exactly the scalars. (**[MEASURED]** On the true Camporesi–Higuchi window the Dirac-commutator seminorm also has kernel exactly the scalars at $n_{\max} \leq 3$, by two independent constructions; the degeneracy reported in earlier versions belongs to a doubled Fock model, Remark 1.9. Whether the convergence can be proved directly in the Dirac seminorm is **[OPEN]**; the proof uses the translation seminorm because its compression estimates close.) The proof is a compression/lifted-state argument in the pattern of the compact-metric-group theorem of Gaudillot-Estrada–van Suijlekom [7], with self-contained proofs: the dual map evaluates conjugated operators against the central spectral Fejér vector state, which embeds *exactly* into the spinor window via the $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ block decomposition $V_j \otimes (V_{j\pm 1/2})$ of the Camporesi–Higuchi eigenspaces (the truncated Hilbert space is the span of complete eigenspaces; §2.2), and both almost-inverse defects collapse to Fejér smoothing — one on functions, one as operator conjugation-averaging — at the common moment $\gamma_{n_{\max}}$. The contribution relative to the scalar theory: the spinor-window transport, per-band injectivity of the compression on the Camporesi–Higuchi window (bands $N \leq 2n_{\max}$), the closed-form rate moment, and the $4/\pi$ asymptotic constant. Supporting structure is organized in five lemmas (the fifth since withdrawn 2026-09-03): (L1′) **[PANEL-VERIFIED]** a finite-distance statement on a doubled computational model of the truncated operator system (not used by the theorem; §2.2); (L2) **[INTERNAL THEOREM + MEASURED]** a positive central spectral Fejér kernel on SU(2) with closed-form mass-concentration moment and Stein–Weiss-type quantitative rate $\gamma_n \sim (4/\pi) \log n/n$ (dual-Coxeter normalisation; $2/\pi$ on the unit S^3); (L3) **[INTERNAL THEOREM]** a Lipschitz comparison bound $\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} \leq C_3 \|\nabla f\|_{L^\infty}$ with $C_3 = 1$, by compression from the continuum identity (proof replaced 2026-09-28; *not* used by the unconditional theorem); (L4) a Berezin map (convolution form) that is positive, contractive, an approximate identity, and L3-compatible; (L5, since *withdrawn* 2026-09-03: the bound is established by the unconditional lifted-state theorem, not this approximation-pair assembly). **[OBSERVATION]** The asymptotic constant is a quotient of sphere volumes — $4/\pi$ in the dual-Coxeter normalisation of §2.1, $2/\pi = \mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ at unit radius, and *not* the Hopf fibration's

*Independent researcher. jloutey@gmail.com. This work was developed in an AI-augmented agentic workflow with Anthropic's Claude. Implementation, internal memos, and bibliographic collation were performed collaboratively; all theorems, design choices, and editorial decisions are the author's responsibility.

base-to-total ratio (§3.2). The $\log n_{\max}$ factor is the first-moment logarithm of the Fejér-type (squared-Dirichlet; on the circle, the Cesàro-1 Fejér) kernel, which the circle's Fejér kernel already carries (Remark 3.5); it is a property of the kernel, not a non-abelian slow-down. State-space Gromov–Hausdorff convergence of scalar truncations for compact metric groups with bi-invariant metric — including SU(2) — is established by Gaudillot-Estrada–van Suijlekom [7]; the residual contribution here is the Camporesi–Higuchi spinor substrate, the closed-form rate moment, and the explicit $4/\pi$ asymptotic constant. The construction supplies quantitative convergence data toward the continuum target S^3, the commutative factor that almost-commutative spectral-triple constructions hosted on S^3 [18] take as input; convergence of an almost-commutative structure itself is not treated here and is **[OPEN]**.

MSC2020 classifications: 58B34 (noncommutative geometry), 46L87 (noncommutative Riemannian and metric geometry), 46L52 (noncommutative function spaces), 22E45 (representations of compact Lie groups).

Keywords: spectral triple, state-space Gromov–Hausdorff convergence, Camporesi–Higuchi Dirac, operator system, Berezin reconstruction, Peter–Weyl, SU(2).

1 Introduction

A central problem in noncommutative metric geometry is whether finite-dimensional truncations of a Riemannian spectral triple converge, in a suitable quantum Gromov–Hausdorff sense, to the continuum geometry. Connes and van Suijlekom [4] introduced the framework of *truncated operator systems* $\mathcal{O}_\Lambda = P_\Lambda \mathcal{A} P_\Lambda$ obtained by spectral projection onto the modes of D with $|D| \leq \Lambda$, and showed that the natural propagation invariant $\mathrm{prop}(\mathcal{O}_\Lambda) = 2$ is realised on the Toeplitz operator system on S^1. They left explicit convergence to the continuum in a quantitative quantum Gromov–Hausdorff distance as an open question, deferring it to "elsewhere." The convergence itself, at the state-space level, is established in van Suijlekom [8] for the circle (spectral and Fourier truncations) and for matrix algebras converging to S^2, and for compact metric groups with bi-invariant metric by Gaudillot-Estrada–van Suijlekom [7]; the residual contribution of the present paper is therefore the spinor substrate, the closed-form rate moment, and the explicit $4/\pi$ constant — not the existence of convergence for SU(2) scalars.

The flat-structure case has been worked out in several subsequent papers. Hekkelman [6] treated the circle S^1 (GH convergence of the pure state spaces of the Toeplitz truncations; state-space convergence for the circle is van Suijlekom's 2021 result [8]). Leimbach and van Suijlekom [17] proved state-space GH convergence of spectral truncations of tori — qualitatively, with no explicit rate — using ball truncations, a spectral Fejér kernel (the square of the spherical Dirichlet kernel) and a Bożejko–Fendler-type transference. A complementary line of work treats *Berezin–Toeplitz* reconstruction on S^2, using the Kähler structure of the coadjoint orbit (see [9] and references therein). All of these results address flat abelian or Kähler structures.

Adjacent activity in the broader propinquity program addresses orthogonal corners of the same general framework. Toyota [13] establishes quantum-Gromov–Hausdorff convergence of spectral truncations for discrete groups of polynomial growth, a scope which excludes compact Lie groups by definition. Farsi–Latrémolière [10] prove spectral continuity for collapse of spectral triples along vertical directions in the spectral propinquity, a collapsing-family setup distinct from the fixed-manifold truncation sequence here. Farsi–Latrémolière [11] establish continuity of the spectral propinquity along analytic paths of Riemannian metrics on a fixed closed spin manifold, a continuous-family setup distinct from the discrete-truncation sequence considered here. Hekkelman–McDonald [12] develop a noncommutative integral and a Szegö limit formula for spectrally truncated triples in the Connes–van Suijlekom paradigm, a companion direction complementary to the metric-side convergence proved here.

For the most natural physically motivated non-Kähler compact case — the round $S^3 = \mathrm{SU}(2)$ — the scalar state-space convergence is covered by the compact-metric-group theorem of

Gaudillot-Estrada–van Suijlekom [7]; what remains distinctive is the *Dirac spectral triple* version on the Camporesi–Higuchi spinor substrate, with explicit closed-form rate.

[INTERNAL THEOREM] The present paper treats that case; the theorem is unconditional under the translation-seminorm metrization of § 1.1. Let $\mathcal{T}_{S^3} = (C^\infty(S^3), L^2(S^3, \Sigma), D_{\mathrm{CH}})$ denote the *Camporesi–Higuchi spectral triple*, with D_{CH} the Dirac operator on the unit round S^3 in the spinor harmonic basis of Camporesi–Higuchi [2], with spectrum $\sigma(D_{\mathrm{CH}}) = \{\pm(n + 1/2) : n \in \mathbb{N}\}$ on each chirality sector.[1] Let $\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, \mathcal{H}_{n_{\max}}, D_{n_{\max}})$ denote its truncation at cutoff $n_{\max}$, where $\mathcal{O}_{n_{\max}}$ is the truncated operator system in the sense of Connes–van Suijlekom and $D_{n_{\max}}$ is the restriction of D_{CH} to $\mathcal{H}_{n_{\max}} = P_{n_{\max}} L^2(S^3, \Sigma)$.

Theorem 1.1 (Main theorem; cf. §1.1 and §4)**.** ***[INTERNAL THEOREM]*** *Unconditionally, the truncated triples $\mathcal{T}_{n_{\max}}$ converge to $\mathcal{T}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance [8], under the translation-seminorm metrization of § 1.1: the bound $\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \leq \gamma_{n_{\max}}$ and $\gamma_{n_{\max}} \to 0$ are proved, where $\gamma_{n_{\max}}$ is the mass-concentration moment of the central spectral Fejér kernel on* SU(2) *given by Lemma 3.4; its displayed asymptotic rate, whose constant is* ***[INTERNAL THEOREM + MEASURED]*** *(Lemma 3.4(d.ii)), is*

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\leq\; \gamma_{n_{\max}}, \qquad \gamma_{n_{\max}} = \frac{4 \log n_{\max}}{\pi\, n_{\max}} + O\big(1/n_{\max}\big) \quad \text{(dual-Coxeter normalisation; } 2/\pi \text{ on the unit } S^3\text{)} \tag{1}$$

The unconditional statement is Theorem 1.7, whose bound is $\leq \gamma_{n_{\max}}$ outright and does not invoke C_3; the main theorem's display now states that bound (until 2026-09-04 it carried a $C_3 = 1$ factor inherited from the withdrawn Lemma 3.12 route). ***[OBSERVATION]*** *The asymptotic constant is $4/\pi$ in the dual-Coxeter normalisation of §2.1 and $2/\pi$ on the unit S^3; the unit-S^3 value is the sphere-volume quotient* $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ *of unit-radius spheres, and the dual-Coxeter value is twice it — though neither is the Hopf base-to-total ratio (see the volume-ratio note in §3.2) (*metric convention, 2026-09-03: *$\gamma_{n_{\max}}$ as computed in Lemma 3.4 weights the kernel by the* SO(3) *rotation angle $\chi \in [0, 2\pi]$, which is* twice *the geodesic distance $\theta = \chi/2$ on the unit round S^3; the constant is $4/\pi$ in that dual-Coxeter normalisation — the round S^3 of radius 2 — and $2/\pi$ on the unit S^3 (the $d_1 = d/2$ metric of §2.1), the two differing by the metric scale, which enters a distance moment linearly; the theorem holds in either normalisation,* `tests/test_p38_metric_convention.py`*).*

The proof structure parallels the abelian Leimbach–van Suijlekom torus argument but requires more machinery on two fronts (a third, listed second below, belonged to earlier versions). First, the central spectral Fejér kernel on SU(2) must be constructed using Peter–Weyl decomposition rather than Schur–Fourier multipliers; the natural-coefficient version (carrying the $\sqrt{2j+1}$ Plancherel weight) is a Fejér-type (squared-Dirichlet) kernel — on the circle, the Cesàro-1 Fejér kernel — whose first moment is of order $\log n/n$ — the logarithm is already present for the circle's Fejér kernel and is a property of the kernel, not a non-abelian slowdown. Second, the Lipschitz comparison estimate on the round S^3 requires the Wen–Avery 3-Y integral [1, 26] to control the multiplier matrices in the truncated basis, and depends on the SO(4) selection rule $|n - n'| + 1 \leq N \leq n + n' - 1$ for non-vanishing 3-Y matrix elements (this was the route of earlier versions; since 2026-09-28 Lemma 3.7 is proved by compression from the continuum identity, and the 3-Y structure enters only supporting computations). Third, the Berezin reconstruction map on SU(2) does not arise from a Kähler structure (the round S^3 has no compatible almost-complex structure), so we use the central-Fejér-kernel spectral form directly, which is the natural non-Kähler analog of van Suijlekom's S^1 Fejér reconstruction [8].

The asymptotic constant in (1) carries geometric content, and its value is normalisation-dependent (2026-09-03): in the dual-Coxeter normalisation of §2.1 the constant is $4/\pi$, which

[1] Camporesi and Higuchi give the spectrum $\pm(n_{\mathrm{CH}} + 3/2)$, $n_{\mathrm{CH}} \geq 0$ [2, §3]; we relabel $n = n_{\mathrm{CH}} + 1$ (a convention of the present work, not of [2]), so the lowest level is $n = 1$ with $|\lambda_1| = 3/2$, matching the SO(4) representation-theoretic indexing used in the supporting computational modules.

coincides numerically with $\mathrm{Vol}(S^2)/\pi^2$ for $\mathrm{Vol}(S^2) = 4\pi$; on the unit S^3 it is $2/\pi$, the factor 2 being the metric scale rather than a geometric signature. *Which volume ratio (2026-09-03).* The constant is a genuine quotient of sphere volumes — but not the one its usual name suggests. For unit-radius spheres $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 4\pi/2\pi^2 = 2/\pi$ exactly, and the dual-Coxeter constant is twice that. It is *not* the Hopf fibration's base-to-total ratio: the Hopf map is a Riemannian submersion $S^3(r) \to S^2(r/2)$, so that ratio is $1/(2\pi)$ on the unit sphere and $1/(4\pi)$ in the dual-Coxeter metric. The *volume content* of the constant — what places it in the $k = 0$ (M1) slot of the master Mellin engine and in $\mathbb{Q}[\pi, \pi^{-1}]$ — is therefore intact; only the "Hopf base" label is a misnomer, that base carrying half the radius of the sphere whose volume appears. Separately, **[OBSERVATION]** the same ratio $2/\pi$ is the first-moment constant of the classical Fejér kernel on the circle (Remark 3.5). This is the volume factor that the $k = 0$ (trivial) heat kernel on the round S^3 leaves behind — the $k = 0$ volume slot of the master Mellin engine, a quotient of unit-sphere volumes rather than the Hopf fibration's base-to-total ratio (§3.2) — and which has been independently identified in the internal supporting analysis as one of the three sub-mechanisms of a master Mellin-engine $\mathrm{Tr}(D^k e^{-tD^2})$ with operator order $k \in \{0, 1, 2\}$ [27]. **[OBSERVATION]** The constant in (1) ($2/\pi$ on the unit S^3; $4/\pi$ in the dual-Coxeter normalisation) sits in the $k = 0$ sub-case (M1, the volume slot) of that classification. (An earlier version said the same $1/\pi^2$ factor reappears in the Pythagorean orthogonalitycorollary (now Paper 42 [21] §11.7, Paper 43 archived); that $1/\pi^2$ is the posited KMS normalisation $\beta = 2\pi/\kappa_g$ squared, not an M1 output, and the link is withdrawn, 2026-09-28.) **[OBSERVATION]** The appearance of this constant in the state-space GH rate is a reading, not a derivation: the rate at which the truncated triples converge to their continuum limit carries the same transcendental signature that the framework's heat-kernel machinery imprints on its bulk observables. No derivation links the two appearances; the shared value is the whole of the evidence.

Outline. Section 2 sets up the round S^3 Camporesi–Higuchi spectral triple, its truncations, and the Peter–Weyl basis. Section 3 states lemmas L1′–L4 and proves or panel-verifies each (tiers inline); Lemma 3.12 is *withdrawn* [retracted 2026-09-05: l5-assembly-listed-as-live-contribution]. Each surviving lemma is self-contained: Lemma 3.1 treats a doubled computational model of the substrate (§2.2); Lemma 3.4 the central spectral Fejér kernel and its quantitative rate; Lemma 3.7 the Lipschitz comparison bound with $C_3 = 1$; Lemma 3.10 the Berezin reconstruction. Section 4 assembles the main theorem via the lifted-state route. Section 5 discusses the role of the $4/\pi$ constant, the Marcolli–van Suijlekom [18] gauge-network framework (for which the result supplies a continuum target rather than an anchor of its diagonal representative), the obstruction to extending the proof to the holomorphic-sector S^5 Bargmann–Segal case, and open questions.

1.1 The unconditional theorem: compression and lifted states under the translation seminorm

Both gaps that conditioned earlier formulations of the main theorem are closed by one reframing: metrize the truncated state space by the *left-translation Lipschitz seminorm* rather than by the truthful-Dirac commutator seminorm. On the continuum the two agree exactly (Lemma 1.3); at finite cutoff the translation seminorm has kernel exactly the scalars at every $n_{\max}$ (Proposition 1.5), and on the true Camporesi–Higuchi window the Dirac seminorm is **[MEASURED]** to have kernel exactly the scalars at $n_{\max} \leq 3$ (Remark 1.9). The translation seminorm is used because its compression and lifted-state estimates close, not because the Dirac seminorm fails; an earlier version of this sentence said the Dirac seminorm degenerates at finite cutoff, which holds only for the doubled Fock model (corrected 2026-09-28). All proofs in this subsection are self-contained; the kernel quantities $K_{n_{\max}}$, $Z_{n_{\max}}$, $\gamma_{n_{\max}}$ are those of § 3.2 below (forward references).

Definition 1.2 (translation seminorm)**.** Let λ_g denote left translation on $S^3 = \mathrm{SU}(2)$, acting on the spinor space $L^2(S^3, \Sigma) \cong L^2(\mathrm{SU}(2)) \otimes \mathbb{C}^2$ (left trivialization) by $U_g = \lambda_g \otimes 1$, and $\rho_g(T) = U_g T U_g^*$. Since D_{CH} is built from the left-invariant frame, $[U_g, D_{\mathrm{CH}}] = 0$; hence U_g preserves every truncation $\mathcal{H}_{n_{\max}}$ and conjugation preserves the operator system: $\rho_g(P_{n_{\max}} M_f P_{n_{\max}}) = P_{n_{\max}} M_{\lambda_g f} P_{n_{\max}}$. For $T \in \mathcal{O}_{n_{\max}}$ and $f \in C(S^3)$ define

$$L_{n_{\max}}(T) := \sup_{g \neq e} \frac{\|\rho_g(T) - T\|_{\mathrm{op}}}{d(e,g)}, \qquad L(f) := \sup_{g \neq e} \frac{\|\lambda_g f - f\|_{L^\infty}}{d(e,g)}, \tag{2}$$

with d the dual-Coxeter geodesic distance of §2.1 — equivalently the rotation angle, $d(e,g) = \chi(g)$ — which is bi-invariant.

Lemma 1.3 (continuum identification)**.** *For $f \in C^\infty(S^3)$, $L(f) = \mathrm{Lip}_d(f) = \|\nabla_d f\|_{L^\infty} = \frac{1}{2}\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}}$, all four quantities taken in the dual-Coxeter metric d of §2.1 except the last, in which D_{CH} is the unit-metric operator (Eq. (6)).*

Proof. $\|\lambda_g f - f\|_\infty = \sup_x |f(g^{-1}x) - f(x)| \leq \mathrm{Lip}(f)\, d(g^{-1}x, x) = \mathrm{Lip}(f)\, d(e,g)$ by bi-invariance, so $L(f) \leq \mathrm{Lip}_d(f)$; conversely left translations act transitively and realize every short geodesic ($d(g^{-1}x, x) = d(e,g)$ along the orbit), so the supremum recovers $\mathrm{Lip}_d(f)$. The identity $\mathrm{Lip}_d(f) = \|\nabla_d f\|_\infty$ on the continuum is standard, as is $\mathrm{Lip}_{d_1}(f) = \|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}}$ for the unit metric (Connes' distance formula on a closed spin manifold); since $d = 2d_1$, Lipschitz seminorms scale inversely and the factor $\frac{1}{2}$ of (6) follows. □

Lemma 1.4 (per-band injectivity)**.** *For $1 \leq N \leq 2n_{\max}$ let $E_N \subset C^\infty(S^3)$ be the degree-$(N-1)$ harmonic band ($\dim E_N = N^2$), an irreducible $\mathrm{SO}(4)$-module of type $V_{(N-1)/2} \otimes V_{(N-1)/2}$. The compression $\iota_N : E_N \to B(\mathcal{H}_{n_{\max}})$, $f \mapsto P_{n_{\max}} M_f P_{n_{\max}}$, is injective.*

Proof. ι_N is $\mathrm{SO}(4)$-equivariant for the conjugation action (rotations implemented on $\mathcal{H}_{n_{\max}}$ commute with $P_{n_{\max}}$). Since E_N is irreducible, ι_N is either zero or injective (Schur). It is nonzero. For $N \leq 2n_{\max} - 1$: by Lemma 1.6(a) every $\phi \otimes \chi$ with ϕ in the scalar window $\bigoplus_{j \leq (n_{\max}-1)/2} V_j \otimes V_j$ lies in $\mathcal{H}_{n_{\max}}$, and $\langle \psi \otimes \chi, M_f\, \phi \otimes \chi \rangle = \int \bar{\psi} f \phi$; the products $\psi\bar{\phi}$ span every band $N \leq 2n_{\max} - 1$ (Clebsch–Gordan series of matrix coefficients), so $\iota_N(f) = 0$ forces $f \perp E_N$, i.e. $f = 0$. For the top band $N = 2n_{\max}$: the block of M_f between the top positive shell $V_{(n_{\max}-1)/2} \otimes V_{n_{\max}/2}$ and the top negative shell $V_{n_{\max}/2} \otimes V_{(n_{\max}-1)/2}$ factors by Wigner–Eckart into a left Clebsch–Gordan coefficient with $J = j_1 + j_2$ and a stretched $6j$ coefficient on the right, both non-zero. Conversely every band $N \geq 2n_{\max} + 1$ compresses to zero: each kept eigenspace $V_j \otimes V_K$ has $j + K \leq n_{\max} - \frac{1}{2}$, so a spin-$J$ multiplier connecting two of them needs $J \leq \min(j_1 + j_2, K_1 + K_2) \leq n_{\max} - \frac{1}{2}$. **[MEASURED]** Two independent constructions of the true window (2026-09-28; the second, Route B, at $n_{\max} \leq 2$) give per-band rank N^2 for every $N \leq 2n_{\max}$, rank 0 above, and $\dim \mathcal{O}_{n_{\max}} = 5, 30, 91$ at $n_{\max} = 1, 2, 3$. (The frozen test `tests/test_p38_action_seminorm.py` checks per-band rank on the doubled computational model of §2.2, whose envelope is $N \leq 2n_{\max} - 1$; it does not see the top band. The true-window freeze is `tests/test_p38_true_ch_window.py`, which pins the top band $N = 2n_{\max}$ together with the rest.) □

Proposition 1.5 (kernel condition on the truthful substrate)**.** *$L_{n_{\max}}(T) = 0$ if and only if $T \in \mathbb{C}1$.*

Proof. $L_{n_{\max}}(T) = 0$ means $\rho_g(T) = T$ for all g. Write $T = P_{n_{\max}} M_f P_{n_{\max}}$ with f band-limited to $N \leq 2n_{\max}$ (every element of $\mathcal{O}_{n_{\max}}$ has this form, bands $N \geq 2n_{\max} + 1$ compressing to zero by Lemma 1.4). Then $P_{n_{\max}} M_{\lambda_g f - f} P_{n_{\max}} = 0$; left translation preserves each harmonic band, so $\lambda_g f - f$ is band-limited and Lemma 1.4 gives $\lambda_g f = f$ for all g; transitivity forces f constant. The converse is immediate. Finite dimensionality plus the kernel condition make $(S(\mathcal{O}_{n_{\max}}), \mathrm{MK}_{L_{n_{\max}}})$ a compact metric space of finite diameter, and Lemma 1.3 provides the same on the continuum side — the standard Monge–Kantorovich setting on both ends. □

Lemma 1.6 (spinor window inclusion and the lifted state). *Let $J = (n_{\max}-1)/2$, let $h = \sum_{j\le J}\sqrt{2j+1}\,\chi_j$ be the central spectral Fejér amplitude of Definition 3.3 (the scalar window then has exactly $n_{\max}$ levels, and $|h|^2/Z_{n_{\max}}$ is the Fejér kernel $K_{n_{\max}}$), fix a unit spinor $\chi\in\mathbb{C}^2$, and set $\xi := (h\otimes\chi)/\sqrt{Z_{n_{\max}}}$. Then:*

*(a) (*Exact window inclusion.*) $\xi\in\mathcal{H}_{n_{\max}}$. Under $\mathrm{SU}(2)_L\times\mathrm{SU}(2)_R$,*

$$L^2(\mathrm{SU}(2))\otimes\mathbb{C}^2 \;=\; \bigoplus_j V_j\otimes\big(V_{j+1/2}\oplus V_{j-1/2}\big),$$

and these summands are precisely the Camporesi–Higuchi eigenspaces [2]: in the level index of (4), $V_j\otimes V_{j+1/2}$ is the positive shell $n=2j+1$ and $V_j\otimes V_{j-1/2}$ the negative shell $n=2j$, with dimensions $n(n+1)$ matching shell by shell. Hence $h\otimes\chi$ lies in the union of the complete *shells $n\le 2J+1=n_{\max}$ — exact containment, no spillover.*

*(b) (*Lifted state.*) For every $f\in C(S^3)$, $\langle\xi,(P_{n_{\max}}M_fP_{n_{\max}})\,\xi\rangle = \int_{S^3} f\,K_{n_{\max}}\,d\mu_{\mathrm{Haar}}$: the induced measure is exactly the Fejér measure, with first moment $\gamma_{n_{\max}}$ (Lemma 3.4(d)).*

*(c) (*Dual map.*) $\upsilon:\mathcal{O}_{n_{\max}}\to C(S^3)$, $\upsilon(T)(g) := \langle U_g\xi,\,T\,U_g\xi\rangle$, is unital and positive, satisfies the tautological contraction $L(\upsilon(T))\le L_{n_{\max}}(T)$, and acts on compressed multipliers as Fejér smoothing: $\upsilon(P_{n_{\max}}M_fP_{n_{\max}}) = K_{n_{\max}}*f$.*

Proof. (a) is the displayed decomposition ($\mathbb{C}^2$ in the right tensor slot; $V_j\otimes V_{1/2}=V_{j+1/2}\oplus V_{j-1/2}$), with the eigenvalue and dimension identifications of Camporesi–Higuchi [2]. (b): $\xi\in\operatorname{ran}P_{n_{\max}}$ absorbs the projections, so the pairing is $\int f\,|h|^2\,|\chi|^2/Z_{n_{\max}} = \int f\,K_{n_{\max}}$. (c): positivity and unitality are immediate from the vector-state form. The contraction: for $q=g'g^{-1}$, $|\upsilon(T)(g)-\upsilon(T)(g')| = |\langle\xi,(\rho_{g^{-1}}(T)-\rho_{g'^{-1}}(T))\,\xi\rangle| \le \|\rho_q(T)-T\|_{\mathrm{op}} \le L_{n_{\max}}(T)\,d(e,q) = L_{n_{\max}}(T)\,d(g,g')$ by bi-invariance. The smoothing identity: $\upsilon(PM_fP)(g) = \int f(x)\,K_{n_{\max}}(g^{-1}x)\,dx = (K_{n_{\max}}*f)(g)$, using centrality and inversion-invariance of $K_{n_{\max}}$; **[MEASURED]** verified to machine precision ($\le 3\times10^{-16}$; the $n_{\max}=2$ scalar window is frozen in `tests/test_p38_action_seminorm.py`, the across-bands sprint check recorded in the CHANGELOG; a spinor-window freeze is owed — the identity itself is proved above). □

Theorem 1.7 (Main theorem, unconditional; **[INTERNAL THEOREM]**). *For every $n_{\max}\ge 1$,*

$$d_{\mathrm{GH}}\Big(\big(S(\mathcal{O}_{n_{\max}}),\mathrm{MK}_{L_{n_{\max}}}\big),\;\big(S(C(S^3)),\mathrm{MK}_L\big)\Big) \;\le\; \gamma_{n_{\max}} \;=\; \frac{4}{\pi}\,\frac{\log n_{\max}}{n_{\max}} + O(1/n_{\max}) \qquad (3)$$

in the dual-Coxeter normalisation of Lemma 3.4, i.e. $\gamma_{n_{\max}} = 2\int K_{n_{\max}}\,d_{\mathrm{round}}\,dg$ on the unit S^3, where the same bound reads $(2/\pi)\log n_{\max}/n_{\max}$ (2026-09-03 convention note; the theorem holds in either normalisation). The inequality and $\gamma_{n_{\max}}\to 0$ are proved; the asymptotic constant is ***[INTERNAL THEOREM + MEASURED]*** *(Lemma 3.4(d.ii), Appendix A).*

Proof. Let $S: C(S^3)\to\mathcal{O}_{n_{\max}}$, $S(f)=P_{n_{\max}}M_fP_{n_{\max}}$ (unital completely positive) and let υ be the dual map of Lemma 1.6. Four facts:

(i) $L_{n_{\max}}(S(f))\le L(f)$: $\rho_g(S(f))-S(f) = S(\lambda_g f - f)$ and compression contracts operator norms.

(ii) $L(\upsilon(T))\le L_{n_{\max}}(T)$ (Lemma 1.6(c)).

(iii) $\|\upsilon(S(f))-f\|_\infty = \|K_{n_{\max}}*f-f\|_\infty \le \gamma_{n_{\max}}\,L(f)$ (the good-kernel estimate in the proof of Lemma 3.10(c), at the function level).

(iv) $\|S(\upsilon(T)) - T\|_{\mathrm{op}} \le \gamma_{n_{\max}} L_{n_{\max}}(T)$: on $T = PM_fP$,

$$S(\upsilon(T)) = P\, M_{K_{n_{\max}} * f}\, P = \int_{\mathrm{SU}(2)} \rho_g(T)\, K_{n_{\max}}(g)\, dg =: \Phi(T)$$

— the operator-side conjugation average against the same Fejér measure — and $\|\Phi(T) - T\|_{\mathrm{op}} \le \int \|\rho_g(T) - T\|_{\mathrm{op}}\, K_{n_{\max}}(g)\, dg \le L_{n_{\max}}(T) \int d(e,g)\, K_{n_{\max}}(g)\, dg = \gamma_{n_{\max}} L_{n_{\max}}(T)$.

By (i)–(ii) the state-space pullbacks $S^* : S(\mathcal{O}_{n_{\max}}) \to S(C(S^3))$ and $\upsilon^* : S(C(S^3)) \to S(\mathcal{O}_{n_{\max}})$ are 1-Lipschitz for the Monge–Kantorovich metrics; by (iii)–(iv) they are $\gamma_{n_{\max}}$-almost mutually inverse, e.g. $d(S^*\upsilon^*\omega, \omega) = \sup_{L(f)\le 1} |\omega(\upsilon(S(f))) - \omega(f)| \le \gamma_{n_{\max}}$. The correspondence $\{(\omega, \upsilon^*\omega)\} \cup \{(S^*\phi, \phi)\}$ then has distortion $\le 2\gamma_{n_{\max}}$ (the standard four-pair approximate-isometry computation), whence $d_{\mathrm{GH}} \le \gamma_{n_{\max}}$. The rate is Lemma 3.4(d.ii)–(d.iii). □

Remark 1.8 (no inverse estimate; the Berezin pair as refinement)**.** Both almost-inverse defects in the proof are Fejér smoothings at the same moment $\gamma_{n_{\max}}$ — one on functions, one as the conjugation average Φ — so no partial-inverse or transference estimate is required anywhere: the dual-reach gap of earlier formulations is dissolved rather than closed. The Berezin map $B_{n_{\max}}$ of Definition 3.9 (Lemmas L4–L5) remains as an explicit forward-reconstruction refinement; it is no longer load-bearing for the distance bound. The scalar $C(G)$ case of this compression/lifted-state pattern, for general compact metric groups with the same single-moment bound, is due to Gaudillot-Estrada–van Suijlekom [7]; the contribution here is the spinor-window transport (the exact-fit inclusion of Lemma 1.6(a)), per-band injectivity of the compression on the Camporesi–Higuchi window, the closed-form rate moment, and the $4/\pi$ asymptotic constant. All assembly steps above are proved self-contained.

Remark 1.9 (the Dirac-commutator seminorm: doubled model versus true window)**. [MEASURED]** *On the true Camporesi–Higuchi window* (complete Dirac eigenspaces up to the cutoff, the object of the main theorem) the kernel of $a \mapsto [D, a]$ on $\mathcal{O}_{n_{\max}}$ is exactly the scalars at $n_{\max} = 1, 2, 3$ ($\dim \mathcal{O}_{n_{\max}} = 5, 30, 91$), so the Dirac-commutator seminorm *is* a norm on $\mathcal{O}_{n_{\max}}/\mathbb{C}\mathbb{1}$ there. Two independent constructions agree, both frozen in `tests/test_p38_true_ch_window.py`: Route A builds the multipliers by Clebsch–Gordan coupling and $D = 2\sigma \cdot J_R + 3/2$ as a matrix (frozen at $n_{\max} = 1, 2, 3$); Route B samples the Wigner-D matrix coefficients with no Clebsch–Gordan code (frozen at $n_{\max} = 1, 2$, where the two routes are asserted to agree band by band). Whether the Gromov–Hausdorff convergence can be proved in this seminorm directly is **[OPEN]**. *On the doubled Fock spin-orbital model* used by the code and by Lemma L1′, the chirality-diagonal commutator seminorm $\|[D_{\mathrm{CH}}, \cdot]\|_{\mathrm{op}}$ has kernel strictly larger than the scalars at the cutoffs measured (10/14 multipliers at $n_{\max} = 2$, 26/55 at $n_{\max} = 3$, bit-exact; every top-band multiplier compresses shell-diagonally), so it does not metrize that model's state space. Earlier versions of this paper read this model degeneracy as a property of the truncated spectral triple and used it to motivate the translation seminorm; that reading is withdrawn (2026-09-28). (The frozen falsifier exercises the $n_{\max} = 2$ counts; the $n_{\max} = 3$ counts are measured but not frozen.) An engineered off-diagonal modification of D_{CH} restores its kernel condition (1/14, 1/55) but is no longer needed for the main theorem. Frozen falsifier for the kernel counts: `tests/test_p45_kplus_degeneracy.py::test_spatial_kernel_condition_truthful_vs_offdiag` (`tests/test_p38_true_ch_window.py` backs the injectivity premise on the true window, and `tests/test_p38_action_seminorm.py` the smoothing identity; neither backs these model kernel counts; pointers corrected 2026-09-03 and 2026-09-30).

Remark 1.10 (history)**.** An earlier draft of this paper (May 2026; repository history and the corresponding Zenodo deposit) stated the main theorem in the Latrémolière propinquity (a formulation since withdrawn), with a Lemma L2 that conflated the kernel's Plancherel mass distribution with its multiplier symbol and a dual-reach argument resting on that conflation. A

verification pass (2026-06-09; repository `CHANGELOG`, v3.106.0) produced an intermediate conditional form; the present unconditional form (2026-06-10) follows from the translation-seminorm reframing of this subsection. (The reframing was motivated by a Dirac-seminorm degeneracy later found to belong to the doubled Fock model only; Remark 1.9.) The companion Lorentzian paper proves the related K^+-compression degeneracy theorem.

2 Setup

Metric convention. The distance in which convergence is proved in this paper is van Suijlekom's *state-space Gromov–Hausdorff distance* [8]: whenever we write $\Lambda(\mathcal{T}_1, \mathcal{T}_2)$ we mean $\mathrm{d}_{\mathrm{GH}}^{\mathrm{vS}}(S(\mathcal{O}_1), S(\mathcal{O}_2))$, the Gromov–Hausdorff distance between the state spaces of the associated operator systems in their Monge–Kantorovich metrics; the definition is restated as (21) in §3.5. The strictly stronger *Gromov–Hausdorff propinquity for metric spectral triples* of Latrémolière [15] (Adv. Math. **404** (2022), Paper No. 108393, preprint `arXiv:1811.10843`, 2018) — the spectral-triple-level extension of the dual Gromov–Hausdorff propinquity of [14] (J. Math. Pures Appl. **103** (2015)) and of the original Gromov–Hausdorff propinquity for quantum compact metric spaces of [16] (Trans. AMS **368** (2016)) — is *not* the distance established here: **[OPEN]** its tunnel hypotheses are not verified for the truncated operator system (§3.5), and convergence in that propinquity remains the strictly stronger open target.

2.1 The round S^3 Camporesi–Higuchi spectral triple

Take $G = \mathrm{SU}(2)$ with the bi-invariant metric in the *dual-Coxeter normalisation*: the Ad-invariant inner product on $\mathfrak{su}(2)$ fixed by $\mathrm{Cas}(\mathrm{ad}) = h^\vee = 2$, namely $\langle X, Y\rangle = -2\,\mathrm{tr}(XY)$ in the defining representation. On SU(2) this normalisation makes the geodesic distance from the identity equal to the *rotation angle*: for $g = \exp(i\theta\,\hat{n}\cdot\sigma)$, which has eigenphases $\pm\theta$, one computes $\|i\theta\,\hat{n}\cdot\sigma\| = 2\theta = \chi(g)$. The underlying manifold is thus the round 3-sphere of radius 2, with diameter 2π and total volume $\mathrm{Vol}(S^3) = 16\pi^2$; the *unit* round S^3 is the same manifold with every distance halved, and we write $d_1 = d/2$ and $\mathrm{Vol}_1(S^3) = 2\pi^2$ when that comparison is needed.

We fix the metric by this representation-theoretic rule, rather than by a choice of radius, because it is the rule under which the rate constant of Theorem 4.1 is comparable across groups: Paper 40 [24] uses the same normalisation for the whole compact connected semisimple class, so the present paper is exactly its rank-1 case. Any uniform rule would do that; this one is the corpus's own convention and *not* the field-standard one: Kac's normalised form $(\theta|\theta) = 2$ [20] gives $\mathrm{Cas}(\mathrm{ad}) = 2h^\vee$, radius $\sqrt{2}$ and constant $2\sqrt{2}/\pi$, so $4/\pi$ is convention-dependent. (Until 2026-09-03 this subsection declared the unit sphere while the moment $\gamma_{n_{\max}}$ of Lemma 3.4 was computed against χ; the two differ by the factor 2 above, and the bound was correspondingly stated with a unit-metric seminorm against a dual-Coxeter moment. Declaring the metric uniformly repairs the pairing in the main theorem: the distance d, the seminorms L and $L_{n_{\max}}$, the moment $\gamma_{n_{\max}}$ and the Monge–Kantorovich distances all scale together (a seminorm inversely, a distance moment directly), so the inequality is unchanged in content — and, since what was printed before paired a *unit*-metric seminorm with a dual-Coxeter moment, the uniform statement is a factor-2 *strengthening* of the printed bound rather than a restatement of equal force. Two scope limits: Lemmas 3.7–3.12 remain stated in the unit metric (Eq. (6) converts), and the panel *height* of §3.5 is scale-*invariant* rather than degree one — a difference of seminorms measured on that seminorm's own unit ball — so the panel comparison there is normalisation-dependent and is not part of the theorem.) Identify S^3 with the Lie group SU(2) via the standard double cover[2].

[2]We use this identification implicitly throughout. It is convenient because (i) the Peter–Weyl decomposition of $L^2(\mathrm{SU}(2))$ provides a natural orthonormal basis indexed by $j \in \frac{1}{2}\mathbb{N}_{\geq 0}$, (ii) the central conjugacy-class measure on SU(2) underlies the mass-concentration analysis of Lemma 3.4, and (iii) the spectrum of D_{CH} has the simplest

Let Σ denote the spinor bundle and $L^2(S^3,\Sigma)$ the spinor Hilbert space.

The Camporesi–Higuchi Dirac operator $D_{\mathrm{CH}} : \mathrm{Dom}(D_{\mathrm{CH}}) \subset L^2(S^3,\Sigma) \to L^2(S^3,\Sigma)$ is the Dirac operator of the *unit* round metric — we keep that normalisation of the operator throughout, so that (4) below is the familiar Camporesi–Higuchi spectrum; the Dirac operator of the dual-Coxeter metric declared above is $\frac{1}{2}D_{\mathrm{CH}}$. It acts on the spinor harmonics

$$D_{\mathrm{CH}}\,\psi_{n,l,m_j,\chi} \;=\; \chi\,(n+\tfrac{1}{2})\,\psi_{n,l,m_j,\chi}, \tag{4}$$

with $n \in \mathbb{N}_{\geq 1}$, $l \in \{0,1,\dots,n-1\}$, $j = l + 1/2$, $m_j \in \{-j, -j+1, \dots, j\}$, and chirality $\chi \in \{+1,-1\}$ [2, §3]. The degeneracy of the level n on each chirality sector is $n(n+1)$; the total degeneracy including both chiralities is $2n(n+1)$. *What the labels are (2026-09-28).* Under $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ the level-n eigenspaces are $V_{(n-1)/2} \otimes V_{n/2}$ ($\chi = +1$) and $V_{n/2} \otimes V_{(n-1)/2}$ ($\chi = -1$) (Lemma 1.6(a)); (j, m_j) label their decomposition under the diagonal $\mathrm{SU}(2)$, j running over $\frac{1}{2}, \dots, n-\frac{1}{2}$, and $l := j - \frac{1}{2}$ is bookkeeping. χ is the sign of the eigenvalue, $\mathrm{sign}(D_{\mathrm{CH}})$ (not a grading, see below).

We then form the Camporesi–Higuchi spectral triple

$$\mathcal{T}_{S^3} \;=\; \big(\,C^\infty(S^3),\; L^2(S^3,\Sigma),\; D_{\mathrm{CH}}\,\big). \tag{5}$$

This is a unital, *odd*, real spectral triple of metric dimension 3 in the sense of Connes [3], with KO-dimension 3 (mod 8). *Parity corrected 2026-09-03:* the text read "even" here, which Connes' table forbids — the grading sign ε'' is defined only in even KO-dimension, so KO-3 carries no chirality, and on S^3 the label $\mathrm{diag}(+1,-1)$ is $\mathrm{sign}(D)$, which *commutes* with D and is therefore not a $\mathbb{Z}_2$ grading. Paper 32 already carries this reading ("KO-3 is the label of the odd continuum GV factor; S^3 carries no chirality") from the 2026-09-02 adjudication; that sweep did not reach this paper. The antiunitary real structure J given by charge conjugation and the Lipschitz seminorm enter as expected (on the truncations J keeps its continuum signs but the order-zero and order-one conditions do not survive; see §2.2). Because D_{CH} is the unit-metric Dirac operator while d is the dual-Coxeter distance, the two differ by the same factor 2:

$$\|f\|_{\mathrm{Lip}} \;:=\; \|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} \;=\; \|\nabla_1 f\|_{L^\infty}, \qquad L(f) \;=\; \mathrm{Lip}_d(f) \;=\; \tfrac{1}{2}\,\|f\|_{\mathrm{Lip}}, \tag{6}$$

so $\|\cdot\|_{\mathrm{Lip}}$ denotes throughout the *unit*-metric Dirac seminorm, while L — the dual-Coxeter Lipschitz seminorm, equivalently the translation seminorm of Definition 1.2 — is what metrizes every state space in the main theorem. Lemmas 3.7–3.12 below are stated in the unit metric (their gradients are the unit-metric ∇_1, as their polar-coordinate form shows), and the conversion between the two is Eq. (6); note that the good-kernel product $\mathrm{Lip}_d(f)\,\gamma_d$ is invariant under the metric scale, since a Lipschitz seminorm and a distance moment scale inversely, so an estimate of that shape is normalisation-independent *provided both factors are taken in the same metric.*

2.2 Truncations

Let $P_{n_{\max}} : L^2(S^3,\Sigma) \to \mathcal{H}_{n_{\max}}$ denote the orthogonal projection onto the finite-dimensional subspace spanned by $\{\psi_{n,l,m_j,\chi} : 1 \leq n \leq n_{\max}\}$. The truncated Hilbert space has dimension

$$\dim \mathcal{H}_{n_{\max}} \;=\; 2\sum_{n=1}^{n_{\max}} n(n+1) \;=\; \tfrac{2}{3}\,n_{\max}(n_{\max}+1)(n_{\max}+2). \tag{7}$$

The truncated Dirac operator is $D_{n_{\max}} = P_{n_{\max}} D_{\mathrm{CH}} P_{n_{\max}}$, which (since D_{CH} is diagonal in the basis above) acts as $\chi(n+1/2)$ on each kept basis vector.

closed form in this convention. We do not draw a structural distinction between S^3 and $\mathrm{SU}(2)$ in this paper.

Substrate: the true window and the computational model (2026-09-28). $\mathcal{H}_{n_{\max}}$ is the span of the *complete* Camporesi–Higuchi eigenspaces with $n \le n_{\max}$, i.e. $|\lambda| \le n_{\max} + \frac{1}{2}$ (Lemma 1.6(a)); it is invariant under left translation. A scalar multiplier $M_f = f \otimes 1$ does *not* preserve $\mathrm{sign}(D_{\mathrm{CH}})$: $P_{n_{\max}} M_f P_{n_{\max}}$ has blocks coupling positive and negative levels (at $n_{\max} = 1$ the whole band $N = 2$ lives in them), and it is non-zero exactly on the bands $N \le 2n_{\max}$ (Lemma 1.4). **[MEASURED]** $\dim \mathcal{O}_{n_{\max}} = 5, 30, 91$ at $n_{\max} = 1, 2, 3$, the sum of N^2 over $N \le 2n_{\max}$ (two independent constructions agree, 2026-09-28; both routes are frozen in `tests/test_p38_true_ch_window.py` — Route A, the matrix construction, at $n_{\max} = 1$–3, and Route B together with the agreement of the two routes at $n_{\max} = 1, 2$ — which also freezes the Dirac spectrum $\pm(n + 3/2)$ with multiplicity $(n+1)(n+2)$ per sign on the same window). The supporting code module and Lemma 3.1 use a different, *doubled* model: two copies of the non-relativistic Fock spin-orbital space $(n, l, j = l + \frac{1}{2}, m_j)$ with $M_f = M_f^{\mathrm{Weyl}} \oplus M_f^{\mathrm{Weyl}}$, Eq. (12). It has the same labels and dimensions, but it is not $\mathrm{SU}(2)_L$-invariant, it omits the sign-mixing blocks, and it reaches only $N \le 2n_{\max} - 1$ (at $n_{\max} = 1$ its operator system is the scalars). The main theorem uses only the true window: Lemma 1.6, Lemma 1.4 and Proposition 1.5 are statements about it, and Theorem 1.7 is unaffected by the model. **[MEASURED]** *Real structure on the true window* (2026-09-28; `tests/test_true_window_real_structure.py`, two independent constructions agreeing pair by pair to 10^{-10} at $n_{\max} = 1, 2$). In the left-invariant frame, spinors $L^2(\mathrm{SU}(2)) \otimes \mathbb{C}^2$ with basis $e^j_{mm's} = \sqrt{2j+1}\, D^j_{mm'} \otimes e_s$ and $D_{\mathrm{CH}} = 2\sigma \cdot J_R + \frac{3}{2}$, the charge conjugation $J\psi = (i\sigma_y)\overline{\psi}$ acts as $e^j_{mm's} \mapsto (-1)^{m-m'} (i\sigma_y)_{s's}\, e^j_{-m,-m',s'}$ composed with complex conjugation. It commutes with D_{CH}, so it preserves $\mathcal{H}_{n_{\max}}$, and there $J^2 = -1$, $JD_{\mathrm{CH}} = +D_{\mathrm{CH}}J$ (the KO-3 signs) and $J\mathcal{O}_{n_{\max}}J^{-1} = \mathcal{O}_{n_{\max}}$ hold exactly for this closed-form J (test-pinned at $n_{\max} \le 4$; the equivariant antiunitaries form a $2n_{\max}$-dimensional family of block phases, test-pinned at $n_{\max} \le 2$, and the tests check $J\mathcal{O}_{n_{\max}}J^{-1} = \mathcal{O}_{n_{\max}}$ for the closed form only). The truncation is *odd* (measured, $n_{\max} \le 3$), like the continuum: the commutant of $\mathcal{O}_{n_{\max}}$ is the scalars at $n_{\max} \le 3$ (test-pinned; at $n_{\max} = 1, 2$ two further routes in the same file — a sampled-function construction, whose generated unital algebra is all of $B(\mathcal{H}_{n_{\max}})$, and an independent polynomial construction on $\mathbb{R}^4$ — corroborate it), so no $\mathbb{Z}_2$-grading exists; the even grading of Paper 32's chirality-doubled model is a property of that model. The signs are forced: the antiunitaries intertwining D_{CH} and the isometry generators are one phase per irreducible block, all with $J^2 = -1$, and none anticommutes with D_{CH}. The order-zero and order-one conditions $[a, Jb^*J^{-1}] = 0$ and $[[D_{\mathrm{CH}}, a], Jb^*J^{-1}] = 0$ *fail*, from $n_{\max} = 1$, with residuals that grow: on the degree-one generators the operator-norm residual is $1/2, 5/9, 11/16, 19/25$ (order zero) and $3/4, 7/9, 13/16, 21/25$ (order one) at $n_{\max} = 1$–4, increasing toward the generators' sup-norm scale 1. The mechanism is exact: $J(PfP)^*J^{-1} = PfP$, so order zero is $[PfP, PgP] = Pg(1-P)fP - Pf(1-P)gP$, a Toeplitz boundary term that vanishes on every Dirac level more than $\max(N_a, N_b) - 1$ below the top (the shell-difference rule $|n - n'| \le N - 1$), and likewise at order one with f replaced by $c(df)$. These failures belong to the finite truncation and do not touch Theorem 1.7, which uses neither condition; the full audit, with the model comparison, is in Paper 32 [19] (axiom audit, item 6). Quantities measured on the model — Lemma 3.1, the multiplier-rank test of `tests/test_p38_action_seminorm.py`, and the model kernel counts (1/14, 1/55) of Remark 1.9 — are model quantities, not window quantities.

Following Connes–van Suijlekom [4, §3], the *truncated operator system* is

$$\mathcal{O}_{n_{\max}} \;=\; P_{n_{\max}}\, C^\infty(S^3)\, P_{n_{\max}} \;\subset\; B(\mathcal{H}_{n_{\max}}), \tag{8}$$

viewed as a self-adjoint subspace of the matrix algebra $M_{\dim \mathcal{H}_{n_{\max}}}(\mathbb{C})$. As emphasised in [4, §1], this is *not* a C^*-subalgebra of $M_{\dim \mathcal{H}_{n_{\max}}}(\mathbb{C})$: the full C^*-algebra it generates is the full matrix algebra (the "C^*-envelope"), but the operator system $\mathcal{O}_{n_{\max}}$ itself encodes finer structure, including the algebra-side analog of the Lipschitz seminorm. The truncated metric spectral triple is then

$$\mathcal{T}_{n_{\max}} \;=\; (\mathcal{O}_{n_{\max}}, \mathcal{H}_{n_{\max}}, D_{n_{\max}}). \tag{9}$$

2.3 Peter–Weyl basis and the Avery 3-Y integral

The Peter–Weyl decomposition of $L^2(\mathrm{SU}(2))$ gives an orthonormal basis $\{D^j_{m,m'} : j \in \tfrac{1}{2}\mathbb{N}_{\geq 0}, |m|, |m'| \leq j\}$ of Wigner D-matrix elements. Setting $n = 2j + 1$ identifies the Peter–Weyl level with the Fock-shell index used above. We work in the *Wen–Avery hyperspherical harmonic* basis

$$Y^{(3)}_{NLM}(\chi, \theta, \phi) \;=\; \mathcal{N}_{NL}\, \sin^L \chi \cdot C^{L+1}_{N-L-1}(\cos\chi)\, Y_{LM}(\theta, \phi), \tag{10}$$

which is the standard SO(4)-adapted basis of $L^2(S^3)$ in the form given by Wen and Avery [1] and used systematically in the molecular orbital literature; here $\mathcal{N}_{NL} = \sqrt{2^{2L+1} N (N-L-1)!\,(L!)^2/(N+L)!}/\sqrt{\pi}$ is the standard normalisation, C^{L+1}_{N-L-1} is the Gegenbauer polynomial, and Y_{LM} are the standard spherical harmonics on S^2. The Laplace–Beltrami spectrum on S^3 is then $-(N^2-1)$ on the isotypic component spanned by $\{Y^{(3)}_{NLM}\}_{L,M}$.

For $f = \sum_{N\geq 1} \sum_{L=0}^{N-1} \sum_{|M|\leq L} c_{NLM}\, Y^{(3)}_{NLM}$, the multiplier matrix $M_f := P_{n_{\max}} f P_{n_{\max}}$ acts on $\mathcal{H}_{n_{\max}}$ with entries

$$(M_f)_{(n',l',m'_j),\,(n,l,m_j)} \;=\; \sum_{N,L,M} c_{NLM}\, \mathcal{G}^{(3)}_{NLM;nlm_j;n'l'm'_j}, \tag{11}$$

where the Avery 3-Y integral $\mathcal{G}^{(3)}$ is the Clebsch–Gordan- weighted sum of Gegenbauer triple integral and S^2 Gaunt integral factors[3].

The selection rule $|n-n'|+1 \leq N \leq n+n'-1$ (stated for the scalar Fock orbitals) enforces a sparse structure on M_f; the pre-2026-09-28 proof of Lemma 3.7 used it, the present proof does not.

3 The five lemmas

3.1 Lemma L1′: chirality-doubled operator system

This subsection concerns the doubled computational model of §2.2, not the true window. In that model the chirality-doubled bundle $\Sigma = \Sigma_+ \oplus \Sigma_-$ gives the truncated Hilbert space the structure $\mathcal{H}_{n_{\max}} = \mathcal{H}^+_{n_{\max}} \oplus \mathcal{H}^-_{n_{\max}}$, with each chirality block carrying a Weyl-sector multiplier matrix M_f^{Weyl} acting identically on the two blocks. The multiplier M_f is therefore block-diagonal in chirality:

$$M_f \;=\; M_f^{\mathrm{Weyl}} \oplus M_f^{\mathrm{Weyl}} \quad \text{on} \quad \mathcal{H}^+_{n_{\max}} \oplus \mathcal{H}^-_{n_{\max}}. \tag{12}$$

Lemma 3.1 (L1′: doubled-model operator system; **[PANEL-VERIFIED]**). *In the doubled model the operator system inherits from (12) a chirality block-diagonal structure (on the true window it does not; §2.2). The truncated Connes distance*

$$d_{\mathcal{T}_{n_{\max}}}(\varphi, \psi) \;=\; \sup\big\{\, |\varphi(M) - \psi(M)| \;:\; M \in \mathcal{O}_{n_{\max}},\; \|[D_{n_{\max}}, M]\|_{\mathrm{op}} \leq 1 \,\big\} \tag{13}$$

is finite on every cross-shell pair of pure node-evaluation states when (12) is replaced by the chirality-mixing offdiagonal Camporesi–Higuchi extension $D^{\mathrm{offdiag}}_{n_{\max}}$ *obtained by adding the standard E1 ladder couplings between $(n, l, m_j, +)$ and $(n+1, l \pm 1, m_j, -)$.*

[3] In SO(4) the relevant 3-Y integral on S^3 factors as a Gegenbauer triple integral times a standard 3-Y integral on S^2, with the SO(4) selection rule $|n-n'|+1 \leq N \leq n+n'-1$ enforcing the truncation $|n-n'| < N$ on the radial channel and the standard S^2 Gaunt selection rule on the angular channel. This double-selection structure entered the pre-2026-09-28 proof of Lemma 3.7 and is treated as canonical infrastructure here. Closed-form values used in internal numerical verifications appear in the supporting code module `geovac/so4_three_y_integral.py` (R3.1, May 2026).

Proof. The block-diagonal structure of M_f in (12) follows directly from the chirality-blindness of scalar multipliers. To verify finiteness of the Connes distance on cross-shell pairs, we observe that a state pair $(\varphi_{n,l,m_j}, \psi_{n',l',m'_j})$ with $n \neq n'$ has finite truncated Connes distance under the offdiag extension because the $E1$ coupling supplies a non-trivial structural multiplier that bounds the supremum in (13). **[PANEL-VERIFIED]** Numerical verification at $n_{\max} \in \{2, 3\}$ shows every cross-shell pair has finite distance, with the four-point Pearson correlation between distance and Fock-graph hop distance taking the value -0.2501 at $n_{\max} = 3$; no frozen test in the repository backs this correlation value.[4]

The role of the offdiag extension is purely SDP-bounding: it ensures that the operator-norm constraint $\|[D_{n_{\max}}, M]\|_{\mathrm{op}} \leq 1$ is non-degenerate on the doubled model's operator system (on the true window the truthful Dirac already is, measured at $n_{\max} \leq 3$, Remark 1.9). The truthful D_{CH} (without offdiag couplings) is what underlies the spectral-action structure of $\mathcal{T}_{S^3}$ and is what we use in Lemmas 3.7–3.10. The main theorem does not use a Dirac-commutator metrization, so this choice does not enter it. □

Remark 3.2. The offdiag extension does *not* anticommute with the chirality-doubling grading $\gamma_{\mathrm{GV}} = \sigma_x \otimes \mathbb{1}$ of the doubled finite model (Paper 32 [19]), whereas the truthful D_{CH} does (`geovac/chirality_grading.py`, `audit_chirality_grading`). That grading exists only on the doubled model: on the continuum S^3 (odd, KO-dimension 3) there is no grading, and $\mathrm{sign}(D_{\mathrm{CH}})$ commutes with D_{CH} (§2.1). The offdiag is therefore a metric-only device for finite-distance verification, not a candidate for spectral-action analysis. The truthful D_{CH} is what enters (4); the state-space GH convergence proof (Theorem 1.7) uses no Dirac commutator.

3.2 Lemma L2: central spectral Fejér kernel on SU(2)

We construct the central spectral Fejér kernel $K_{n_{\max}} \in C(\mathrm{SU}(2))$ that plays the role of the circle Fejér kernel in van Suijlekom's reconstruction [8] but with the natural-coefficient Plancherel weights forced by the Peter–Weyl bijection $n = 2j + 1$.

Definition 3.3 (Central spectral Fejér kernel)**.** For $n_{\max} \geq 1$, define

$$K_{n_{\max}}(g) := \frac{1}{Z_{n_{\max}}} \left| \sum_{j \leq j_{\max}} \sqrt{2j+1}\, \chi_j(g) \right|^2, \qquad Z_{n_{\max}} = \frac{n_{\max}(n_{\max}+1)}{2}, \tag{14}$$

where χ_j is the trace character of the $(2j+1)$-dimensional irreducible representation of SU(2) and $j_{\max}$ is the largest half-integer with $2j_{\max} + 1 \leq n_{\max}$.

The kernel is positive on SU(2) (manifestly), central (depends only on the conjugacy class of g), and unit-mass under the standard SU(2) Haar measure dg. Identifying conjugacy classes with the rotation angle $\chi \in [0, 2\pi]$ via $g = \exp(i(\chi/2)\, \hat{n} \cdot \sigma)$ on the maximal torus (eigenphases $\pm\chi/2$; unit-S^3 geodesic distance $\chi/2$; corrected 2026-09-03 from $\chi \in [0, \pi]$ with $g = \exp(i\chi \hat{n} \cdot \sigma)$, which is the geodesic angle and does not match the characters below), the conjugacy-class measure is $d\mu(\chi) = (1/\pi) \sin^2(\chi/2)\, d\chi$.

Lemma 3.4 (L2: kernel construction and quantitative rate)**.** *The central spectral Fejér kernel $K_{n_{\max}}$ has the following properties (the closed-form $\gamma_{n_{\max}}$ rate moment is verified by `tests/test_central_fejer` and the $4/\pi$ asymptotic constant by `tests/test_trunk_qa_fejer_4_over_pi.py`; $2/\pi$ is the same constant on the unit S^3 and the circle's constant, not a decoy):*

(a) $K_{n_{\max}}(g) \geq 0$ *for all* g, *and* $\int_{\mathrm{SU}(2)} K_{n_{\max}}(g)\, dg = 1$.

[4] The numerical correlation value is recorded in the internal computation log; the structural finiteness was used by the withdrawn Lemma 3.12 route, and is robust under operator-norm perturbations of D that preserve the SU(2) selection rules.

(b) *(Mass distribution and multiplier symbol.) The kernel's Plancherel* mass distribution *is* $m_{n_{\max}}(j) = (2j+1)/Z_{n_{\max}}$ *on* $j \le j_{\max}$ *(the squared Plancherel weights of the amplitude* $\sum_j \sqrt{2j+1}\,\chi_j$*), a probability distribution with maximum* $2/(n_{\max}+1)$ *at* $j_{\max}$*. The convolution operator* $T_{K_{n_{\max}}} : f \mapsto K_{n_{\max}} * f$ *acts on the* j'*-isotypic component by the* multiplier symbol $\sigma_{n_{\max}}(j') = c_{j'}/((2j'+1)Z_{n_{\max}})$ *with* $c_{j'} = \sum_{j_1,j_2 \le j_{\max},\, \triangle(j_1,j_2,j')} \sqrt{(2j_1+1)(2j_2+1)}$*, where* $\triangle$ *is the Clebsch–Gordan condition* with unit steps *(*$|j_1 - j_2| \le j' \le j_1 + j_2$ *and* $j_1 + j_2 + j' \in \mathbb{Z}$*):* $\sigma_{n_{\max}}(0) = 1$*,* $0 \le \sigma_{n_{\max}} \le 1$*, support* $j' \le 2j_{\max}$*.* ***[MEASURED]*** *At* $n_{\max} = 2$*:* $\sigma = (1, \sqrt{2}/3, 2/9)$ *at* $j' = (0, \frac{1}{2}, 1)$ *(bit-exact against independent Weyl-integration quadrature).*

(c) *(Cb-norm.)* $T_{K_{n_{\max}}}$ *is unital completely positive (convolution against a probability density), hence* $\|T_{K_{n_{\max}}}\|_{\mathrm{cb}} = 1$*. (The mass-distribution maximum* $2/(n_{\max}+1)$ *of (b) is* not *the cb-norm of any map used in this paper.)*

(d) *(Mass-concentration moment.) Define*

$$\gamma_{n_{\max}} := \int_{\mathrm{SU}(2)} K_{n_{\max}}(g)\,\chi(g)\,dg, \tag{15}$$

where $\chi(g) \in [0, 2\pi]$ *is the class (rotation) angle of* g*, which by §2.1* is *the dual-Coxeter geodesic distance* $d(e,g)$ *(*$g = \cos\theta + i \sin\theta\, \hat{n}\cdot\sigma$ *has eigenphases* $\pm\theta$ *and* $d(e,g) = 2\theta = \chi$*; the characters below are* $\sin((2j+1)\chi/2)/\sin(\chi/2)$*). So* $\gamma_{n_{\max}}$ *is the mass-concentration moment against the paper's own metric, and every constant below is a dual-Coxeter constant; the corresponding unit-*S^3 *moment is exactly* $\gamma_{n_{\max}}/2$ *at every* $n_{\max}$ *(*`tests/test_p38_metric_conve`
$K_1 \equiv 1$ makes $\gamma_1 = \pi$, exactly twice the unit-S^3 Haar-mean distance $\pi/2$). Correction (2026-09-03): *this definition previously wrote "*d_{round}*" while §2.1 declared the unit sphere, so the moment and the seminorm were stated in different metrics; both are now dual-Coxeter. Then:*

(d.i) ***[INTERNAL THEOREM]*** $\gamma_{n_{\max}}$ *admits the closed-form sum-rule*

$$\gamma_{n_{\max}} = \pi - \frac{4T_{n_{\max}}}{\pi\, Z_{n_{\max}}}, \qquad T_{n_{\max}} := \sum_{\substack{1 \le k_1, k_2 \le n_{\max} \\ k_1 + k_2 \text{ odd}}} \sqrt{k_1 k_2}\left[\frac{1}{(k_1 - k_2)^2} - \frac{1}{(k_1 + k_2)^2}\right]. \tag{16}$$

(d.ii) *(Asymptotic rate.)* ***[INTERNAL THEOREM + MEASURED]*** $\lim_{n_{\max}\to\infty} \dfrac{n_{\max}\,\gamma_{n_{\max}}}{\log n_{\max}} = \dfrac{4}{\pi}$.

(d.iii) *(Uniform bound.)* ***[PANEL-VERIFIED]*** $\gamma_{n_{\max}} \le 6\,\log n_{\max}/n_{\max}$ *for all* $n_{\max} \ge 2$ *(established through a monotonicity statement checked on a finite range of* $n_{\max}$*, not proved for all* $n_{\max}$*).*

Proof sketch. Property (a) is a standard Peter–Weyl computation. For (b), the mass distribution is immediate from the amplitude weights; the multiplier symbol follows by expanding $|h|^2$ with the SU(2) Clebsch–Gordan rule $\chi_{j_1}\chi_{j_2} = \sum_{J=|j_1-j_2|}^{j_1+j_2} \chi_J$ (unit steps in J, so $j_1 + j_2 + J \in \mathbb{Z}$) and reading off the convolution eigenvalue $c_{j'}/((2j'+1)Z)$ on each block; unitality at $j' = 0$ is $c_0 = \sum_j (2j+1) = Z$, and the $n_{\max} = 2$ values are verified bit-exactly against independent Weyl-integration quadrature (supporting computation in the project repository, `debug/p45_adversarial_L2_memo.md`). Property (c): a unital positive map on a commutative domain is completely positive, and a UCP map has cb-norm exactly 1; on the truncated side the induced central multiplier acts block-scalar on $\bigoplus_\pi M_{d_\pi}(\mathbb{C})$ with cb-norm $\sup_\pi |\sigma(\pi)| = 1$ — an elementary statement requiring no amenability input.

For (d), the closed form (d.i) follows from rewriting the kernel using $\sin a \sin b = \frac{1}{2}(\cos(a-b) - \cos(a+b))$ and applying the elementary integral $\int_0^{2\pi} \chi \cos(m\chi/2)d\chi = -8/m^2$ for odd

$m \neq 0$, $2\pi^2$ for $m = 0$, and 0 for non-zero even m. Substituting and separating diagonal from off-diagonal contributions gives (16).

For (d.ii), expand $\sqrt{k_1 k_2} = \sqrt{a(a+d)} \approx a + d/2 + O(d^2/a)$ for $k_2 = k_1 + d$, d odd, $a \gg d$. The leading-order sum is $\sum_{d\,\mathrm{odd}} 1/d^2 \cdot \sum_a a \to (\pi^2/8) \cdot n^2/2$, giving $T_n/Z_n \to \pi^2/4$ and $\gamma_n \to 0$. The sub-leading $\log n/n$ correction is the net of two logarithmic pieces — the triangle truncation of the odd-d sum (coefficient 2) and the $d/2$ term of $\sqrt{a(a+d)}$ (coefficient -1) — computed in Appendix A, Step 3:

$$\frac{T_n}{Z_n} \;=\; \frac{\pi^2}{4} \;-\; \frac{\log n}{n} \;+\; O(1/n),$$

which gives $\gamma_n = (4/\pi)\log n/n + O(1/n)$.

For (d.iii), **[PANEL-VERIFIED]** the function $n \mapsto n\gamma_n/\log n$ is monotonically decreasing for $n \geq 3$ (verified on the sample $n \in \{3, 4, 5, 7, 10, 15, 20, 30, 50, 100, 200, 500, 1000\}$ by the frozen test `tests/test_central_fejer_su2.py`; no proof of monotonicity for all n is given), with maximum at $n = 2$ equal to $2\gamma_2/\log 2 = 2(\pi - 64\sqrt{2}/(27\pi))/\log 2 \approx 5.986$. The uniform bound follows. □

Remark 3.5 (Connection to the circle Fejér estimate)**.** The asymptotic rate $\gamma_n \sim (4/\pi)\log n/n$ is the non-abelian analog of the classical Fejér-on-the-circle estimate

$$\frac{1}{2\pi}\int_{\mathbb{T}^1} F_n(\theta)\,|\theta|\,d\theta \;\sim\; \frac{2}{\pi}\,\frac{\log n}{n}, \qquad F_n(\theta) = \frac{1}{n}\,\frac{\sin^2(n\theta/2)}{\sin^2(\theta/2)}, \tag{17}$$

for the probability-normalised Fejér kernel ($\frac{1}{2\pi}\int_{\mathbb{T}^1} F_n = 1$; Zygmund [30] Vol. I, Ch. III). The circle moment has the exact closed form $\frac{1}{2\pi}\int_{\mathbb{T}^1} F_n|\theta| = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k\ \mathrm{odd},\,k<n}(1 - k/n)/k^2$ (from $\int_0^\pi \theta\cos k\theta\,d\theta = ((-1)^k - 1)/k^2$), whose $\log n/n$ coefficient is $2/\pi$ (the $\frac{1}{n}\sum_{k\ \mathrm{odd}} 1/k \sim \frac{1}{2n}\log n$ term). **[SYMBOLIC + MEASURED]** The circle constant is $2/\pi$, not $4/\pi$: the doubling estimator $(2n\,m_{2n} - n\,m_n)/\log 2$ on the closed form gives 0.63662 at $n = 400$ ($2/\pi = 0.63662$), the SU(2) doubling estimator of Lemma 3.4(d.ii), $a_n = (2n\gamma_{2n} - n\gamma_n)/\log 2$, reads 1.2785 at $n = 800$ and 1.2761 at $n = 1600$ (i.e. using γ_{1600} and γ_{3200}; three digits of $4/\pi = 1.2732$, approaching from above with the residual shrinking by $\sim$1.85× per doubling; Appendix A), and the ratio of the two estimators is 2.008 at $n = 800$ (`tests/test_trunk_qa_fejer_4_over_pi.py`: `test_circle_fejer_closed_form_matches_quadrature`, `test_circle_fejer_constant_is_2_over_pi`, `test_su2_constant_is_twice_circle`). An earlier version of this remark stated the circle constant as $4/\pi$, "the same on both sides"; that statement is withdrawn (2026-09-02 certification run). **[OBSERVATION]** In the dual-Coxeter normalisation the SU(2) constant is twice the circle constant; in the unit-S^3 normalisation (geodesic distance, the circle's own convention) the two coincide at $2/\pi$ (circle: $2\mathrm{Vol}(S^0)/\mathrm{Vol}(S^1) = 4/(2\pi) = 2/\pi$; SU(2): $\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2)) = 4\pi/(2\pi^2) = 2/\pi$ — the same number, reached with a factor 2 on the circle side and without it on the SU(2) side, so no common "2Vol(base)/Vol(group)" form is claimed), the ratio of the unit-metric SU(2) estimator to the circle's being 1.004 at $n = 800$ (2026-09-03 correction: the factor 2 reported before is the metric scale, not a group-theoretic signature). No derivation of the SU(2) constant from the circle constant is given here; the $\sin^2(\chi/2)$ conjugacy-class weight and the $\sqrt{2j+1}$ Plancherel weight of Definition 3.3 together change both the kernel and the constant, while preserving the leading $\log n/n$ behaviour that the Fejér (on the circle, Cesàro-1) first moment already has there. The $\log n$ factor is therefore a property of the kernel, not a non-abelian slowdown. (An earlier version compared it with a torus rate $O(1/n_{\max})$ attributed to Leimbach–van Suijlekom [17]; that paper proves convergence without an explicit rate, and the comparison is withdrawn, 2026-09-28.)

Remark 3.6 (The volume ratio behind the constant)**. [OBSERVATION]** In the dual-Coxeter normalisation the asymptotic constant coincides numerically with

$$\frac{4}{\pi} \;=\; \frac{\mathrm{Vol}(S^2)}{\pi^2} \;=\; \frac{4\pi}{\pi^2}.$$

Which volume ratio (2026-09-03). The constant is a genuine quotient of sphere volumes — but not the one its usual name suggests. For unit-radius spheres $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 4\pi/2\pi^2 = 2/\pi$ exactly, and the dual-Coxeter constant is twice that. It is *not* the Hopf fibration's base-to-total ratio: the Hopf map is a Riemannian submersion $S^3(r) \to S^2(r/2)$, so that ratio is $1/(2\pi)$ on the unit sphere and $1/(4\pi)$ in the dual-Coxeter metric. The *volume content* of the constant — what places it in the $k = 0$ (M1) slot of the master Mellin engine and in $\mathbb{Q}[\pi, \pi^{-1}]$ — is therefore intact; only the "Hopf base" label is a misnomer, that base carrying half the radius of the sphere whose volume appears. Read as the unit-sphere quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ rather than as a Hopf base-to-total ratio, the constant still carries pure volume content, and is one of the natural transcendentals on SU(2) that survives at the leading order of the central Fejér kernel's first moment.

3.3 Lemma L3: Lipschitz comparison with $C_3 = 1$

The next lemma is the Lipschitz comparison estimate that controls the operator norm of the commutator $[D_{\mathrm{CH}}, M_f]$ in terms of the classical Lipschitz norm of f on the round S^3.

Lemma 3.7 (L3: Lipschitz comparison, truthful D_{CH} on S^3; **[INTERNAL THEOREM]**). *For every $f \in C^\infty(S^3)$ and every $n_{\max} \geq 1$,*

$$\|[D_{\mathrm{CH}},\, M_f]\|_{\mathrm{op}} \;\leq\; C_3 \cdot \|\nabla f\|_{L^\infty(S^3)}, \qquad C_3 = 1, \tag{18}$$

where the round-S^3 Lipschitz norm $\|\nabla f\|_{L^\infty}^2 = (\partial_\chi f)^2 + \sin^{-2}\chi\,(\partial_\theta f)^2 + \sin^{-2}\chi \sin^{-2}\theta\,(\partial_\phi f)^2$ is taken in the standard polar parametrisation.

Panel check (on the doubled model of §2.2): `tests/test_r25_l3_lipschitz_bound.py` (`test_eq_l3_main_ineq`, `test_l3_C3_uniform_in_n_max`).

Proof. Here $M_f = P_{n_{\max}} M_f P_{n_{\max}}$. Since $[D_{\mathrm{CH}}, P_{n_{\max}}] = 0$, $[D_{\mathrm{CH}}, M_f] = P_{n_{\max}}[D_{\mathrm{CH}}, M_f]P_{n_{\max}}$, and compression contracts operator norms, so $\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} \leq \|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} = \|\nabla f\|_{L^\infty}$, the last equality being the continuum identity (6) (Connes' formula on a closed spin manifold: $[D_{\mathrm{CH}}, M_f]$ is Clifford multiplication by df). This holds for every f and every $n_{\max}$. □

Remark 3.8 (the constant is sharp in the limit; earlier proof withdrawn). Since $P_{n_{\max}} \to 1$ strongly, $\|P_{n_{\max}}[D_{\mathrm{CH}}, M_f]P_{n_{\max}}\|_{\mathrm{op}} \to \|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}}$, so $C_3 = 1$ cannot be lowered uniformly in $n_{\max}$. Until 2026-09-28 this lemma carried a per-harmonic proof asserting the ratio $\|[D_{\mathrm{CH}}, M_{NLM}]\|_{\mathrm{op}}/\|\nabla Y^{(3)}_{NLM}\|_{L^\infty} \leq \sqrt{(N-1)/(N+1)} < 1$ at every cutoff, with per-cutoff values and a "tight because of SO(4) sparsity" reading. That inequality is false (for $N = 2$ the ratio tends to 1, not $1/\sqrt{3}$, by the limit just stated), two of its steps ran the wrong way, and the per-cutoff values and the sparsity reading are withdrawn with it. The conclusion $C_3 = 1$ stands on the compression proof above.

3.4 Lemma L4: Berezin reconstruction

Definition 3.9 (Berezin map, convolution form). For $n_{\max} \geq 1$ and $f \in C^\infty(S^3)$, define

$$B_{n_{\max}}(f) \;:=\; P_{n_{\max}}\, M_{K_{n_{\max}} * f}\, P_{n_{\max}} \;=\; \sum_{N \leq 2n_{\max}-1} \sum_{L=0}^{N-1} \sum_{|M| \leq L} \sigma_{n_{\max}}(N)\, c_{NLM}\, M_{NLM}, \tag{19}$$

where $\sigma_{n_{\max}}(N)$ is the multiplier symbol of Lemma 3.4(b) under the bijection $n = 2j + 1$ and $f = \sum c_{NLM} Y^{(3)}_{NLM}$. (A sum form weighted by the mass distribution $N/Z_{n_{\max}}$ in place of $\sigma_{n_{\max}}(N)$ would not be unital — $B(\mathbf{1}) = Z^{-1} I$ — and would fail the approximate-identity property maximally at $f = \mathbf{1}$. Note the image reaches multiplier labels up to $N \leq 2n_{\max} - 1$, the support of $\sigma_{n_{\max}}$ — one band below the window's envelope $N \leq 2n_{\max}$, Lemma 1.4.)

Lemma 3.10 (L4: properties of the Berezin reconstruction)**.** *The map $B_{n_{\max}} : C^\infty(S^3) \to \mathcal{O}_{n_{\max}}$ has the following four properties:*

(a) (Positivity.) If $f \geq 0$ on S^3, then $B_{n_{\max}}(f)$ is positive semi-definite.

(b) (Contractivity.) $\|B_{n_{\max}}(f)\|_{\mathrm{op}} \leq \|f\|_{L^\infty}$.

(c) (Approximate identity.) $\|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\mathrm{op}} \leq \gamma_{n_{\max}} \cdot \|\nabla f\|_{L^\infty}$.

(d) (Compatibility with L3.) $\|[D_{\mathrm{CH}}, B_{n_{\max}}(f)]\|_{\mathrm{op}} \leq \|\nabla f\|_{L^\infty}$ (via Lemma 3.7).

Proof. **(a)** Equivalent to the convolution form $B_{n_{\max}}(f) = P_{n_{\max}}(K_{n_{\max}} * f)P_{n_{\max}}$: since $K_{n_{\max}} \geq 0$ pointwise (Lemma 3.4(a)) and the convolution of two non-negative functions on a compact group is non-negative, $K_{n_{\max}} * f \geq 0$. Compression by an orthogonal projection preserves positivity: for any $\psi \in \mathcal{H}_{n_{\max}}$,

$$\langle \psi | P_{n_{\max}} M_g P_{n_{\max}} | \psi \rangle \;=\; \int_{S^3} g(\omega)\, |P_{n_{\max}} \psi(\omega)|^2 \, d\Omega \;\geq\; 0.$$

(b) By Young's inequality, $\|K_{n_{\max}} * f\|_{L^\infty} \leq \|K_{n_{\max}}\|_{L^1} \|f\|_{L^\infty} = \|f\|_{L^\infty}$. Compression contracts operator norms, so $\|B_{n_{\max}}(f)\|_{\mathrm{op}} \leq \|f\|_{L^\infty}$.

(c) The standard Lipschitz/good-kernel estimate gives

$$\|K_{n_{\max}} * f - f\|_{L^\infty} \;\leq\; \int_{\mathrm{SU}(2)} K_{n_{\max}}(g)\, |f(g \cdot x) - f(x)|\, dg \;\leq\; \gamma_{n_{\max}}\, \|\nabla f\|_{L^\infty},$$

which is the SU(2) version of the standard approximate-identity estimate (Stein–Weiss [29], Ch. I). Compression then gives $\|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\mathrm{op}} \leq \gamma_{n_{\max}} \|\nabla f\|_{L^\infty}$.

(d) The cleanest argument goes via the convolution form $B_{n_{\max}}(f) = P_{n_{\max}}\, M_{K_{n_{\max}} * f}\, P_{n_{\max}}$ used in parts (a) and (c). Let $g := K_{n_{\max}} * f \in C^\infty(S^3)$. Since $K_{n_{\max}} \geq 0$ and $\|K_{n_{\max}}\|_{L^1} = 1$ by Lemma 3.4(a), Young's inequality for the gradient

$$\|\nabla g\|_{L^\infty} \;=\; \|\nabla (K_{n_{\max}} * f)\|_{L^\infty} \;=\; \|K_{n_{\max}} * \nabla f\|_{L^\infty} \;\leq\; \|K_{n_{\max}}\|_{L^1}\, \|\nabla f\|_{L^\infty} \;=\; \|\nabla f\|_{L^\infty} \quad (20)$$

gives a smoothed function g whose Lipschitz norm is no larger than that of f (left-translation by $K_{n_{\max}}$ commutes with the gradient on the bi-invariant metric and convolution with a probability density does not increase the Lipschitz norm). Now apply Lemma 3.7 to g in the truncated multiplier representation: $B_{n_{\max}}(f) = M_g$ in $\mathcal{O}_{n_{\max}}$, so

$$\|[D_{\mathrm{CH}}, B_{n_{\max}}(f)]\|_{\mathrm{op}} \;=\; \|[D_{\mathrm{CH}}, M_g]\|_{\mathrm{op}} \;\leq\; C_3\, \|\nabla g\|_{L^\infty} \;\leq\; C_3\, \|\nabla f\|_{L^\infty} \;=\; \|\nabla f\|_{L^\infty},$$

where the final equality uses $C_3 = 1$ from Lemma 3.7. □

Remark 3.11 (Non-Kähler analog)**.** The Berezin reconstruction (19) differs from the standard Hawkins [5] construction on Kähler manifolds, which proceeds via the holomorphic Toeplitz form on a Hardy space. Since $S^3 = \mathrm{SU}(2)$ admits no compatible almost-complex structure (it is parallelisable but not Kähler), the natural analog is the central-spectral-Fejér-kernel form, which is the compact-group (class-function) version of the same idea applied to the conjugacy-class measure on SU(2). The Kähler analog on $S^2 = \mathrm{SU}(2)/U(1)$ via Berezin–Toeplitz on the Hopf base has been worked out independently (see [9] and references therein); we have not checked its relation to the present construction.

Panel note (2026-09-03). The reference implementation (`geovac/gh_convergence.py`) reports reach and height on a fixed test panel of low-degree harmonics. Those panel numbers do

not verify this lemma: the Lipschitz-normalised panel height rises with the cutoff while $\gamma_{n_{\max}}$ falls, and the two cross between $n_{\max} = 6$ and $n_{\max} = 7$ (measured 2026-09-03):

$n_{\max}$	2	3	4	5	6	7
$\gamma_{n_{\max}}$	2.075	1.610	1.322	1.130	0.990	0.883
height/$\|f\|_{\mathrm{Lip}}$	0.667	0.808	0.878	0.913	0.937	0.952
margin	+1.408	+0.803	+0.445	+0.217	+0.053	−0.069

Normalised as printed, the panel entries rescale into the unit-Lipschitz ball, so they are a subset of the set quantified over and a lower bound on the supremum: an entry exceeding $\gamma_{n_{\max}}$ contradicts the withdrawn bound. The analytic witness of (24) settles the question independently of the panel. The proof below does not use the panel; identifying the panel-side quantity is an open check.

3.5 Lemma L5: state-space Gromov–Hausdorff assembly

A remark on framework before assembling. In Latrémolière's Gromov–Hausdorff propinquity for metric spectral triples [15], the distance is an infimum over tunnels carried by quantum metric spaces with quantum isometries, under hypotheses (Leibniz Lip-norms on C*-algebras; kernel condition) not established for the truncated operator system; a pair of UCP maps is not a tunnel there. The distance the pair $(B_{n_{\max}}, P_{n_{\max}})$ certifies is van Suijlekom's state-space Gromov–Hausdorff distance [8], via two-directional approximation data in the Leimbach–van Suijlekom pattern [17]. We write

$$\Lambda(\mathcal{T}_1, \mathcal{T}_2) \;:=\; \mathrm{d}_{\mathrm{GH}}^{\mathrm{vS}}\big(S(\mathcal{O}_1),\, S(\mathcal{O}_2)\big) \tag{21}$$

for this distance (between state spaces with their respective Monge–Kantorovich metrics), and retain the constituent names reach_B, reach_P, height_B, height_P for the four approximation estimates the pair supplies. The truncated-side Monge–Kantorovich metric is non-degenerate unconditionally under the translation-seminorm metrization (Proposition 1.5); the distance bound itself is Theorem 1.7, and the pair below supplies the explicit forward-reconstruction refinement.

Lemma 3.12 (L5: Berezin-pair refinement of the forward direction; **[REFUTED AS STATED — withdrawn 2026-09-03]**). *Within the framework of § 1.1 (kernel condition Proposition 1.5; distance bound Theorem 1.7), the pair $(B_{n_{\max}}, P_{n_{\max}})$ with $B_{n_{\max}}$ from Definition 3.9 and $P_{n_{\max}}$ the standard spectral projection yields the bound*

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\le\; C_3 \cdot \gamma_{n_{\max}}, \tag{22}$$

with $C_3 = 1$ (Lemma 3.7) and $\gamma_{n_{\max}}$ from Lemma 3.4(d).

Proof. The four constituents a Latrémolière tunnel would require are:

1. $\mathrm{reach}_B := \sup_{\|f\|_{\mathrm{Lip}} \le 1} \|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\mathrm{op}}$,
2. reach_P, the dual reach via the partial inverse,
3. height_B, the Lipschitz-distortion envelope of $B_{n_{\max}}$ (defined precisely below),
4. $\mathrm{height}_P = 0$ for the orthogonal projection *(withdrawn: false by the same finite-band witness as* height_B*, see below).*

Reach. By L4(c), $\mathrm{reach}_B \le \gamma_{n_{\max}} \cdot \|\nabla f\|_{L^\infty} \le \gamma_{n_{\max}}$ on the unit Lipschitz ball $\{\|f\|_{\mathrm{Lip}} \le 1\}$, where $\|\nabla f\|_{L^\infty} = \|f\|_{\mathrm{Lip}}$ by (6) (an identity; an earlier version appealed to Lemma 3.7's then-panel scope here, which was unnecessary). **(Dissolved.)** The dual estimate is no longer needed: Theorem 1.7 obtains the distance bound via the lifted-state map, whose almost-inverse defect is the conjugation average Φ at the same moment $\gamma_{n_{\max}}$ (Remark 1.8) — no partial inverse of

$B_{n_{\max}}$, and no transference estimate, enters anywhere. (For the record: a partial inverse of $B_{n_{\max}}$ on the full support window would be governed by the symbol *minimum* and grow like $O(n_{\max}^2)$, which is why no forward bound could supply it.) The pair $(B_{n_{\max}}, P_{n_{\max}})$ retains its role as the explicit forward-reconstruction refinement.

Height. The height constituent is a GeoVac-defined approximation quantity, loosely modelled on the height and reach of bridges in [16]; it is not a quantity of the spectral propinquity [15], whose tunnels are built from quantum isometries and so carry no Lipschitz distortion. It compares Lipschitz seminorms rather than operator norms. Specifically,

$$\mathrm{height}_B \;:=\; \sup_{\|f\|_{\mathrm{Lip}}\le 1} \Big|\, \|f\|_{\mathrm{Lip}} - \|B_{n_{\max}}(f)\|_{\mathrm{Lip}}^{\mathcal{O}_{n_{\max}}} \Big|. \tag{23}$$

[REFUTED AS STATED — withdrawn 2026-09-03.] The argument printed here until that date concluded $\mathrm{height}_B \le \gamma_{n_{\max}}$ and $\mathrm{height}_P = 0$. Both are false, by a one-line witness. $B_{n_{\max}}$ is a finite-band reconstruction — its multiplier envelope is $N \le 2n_{\max} - 1$ — so at every finite cutoff the unit-Lipschitz ball contains f with $B_{n_{\max}}(f) = 0$, and for any such f the quantity in (23) equals $\|f\|_{\mathrm{Lip}} = 1$. Hence

$$\mathrm{height}_B \;\equiv\; 1 \quad \text{at every } n_{\max}, \tag{24}$$

and $\mathrm{height}_B \le \gamma_{n_{\max}}$ fails whenever $\gamma_{n_{\max}} < 1$, i.e. for every $n_{\max} \ge 6$ ($\gamma_6 = 0.98958$, $\gamma_7 = 0.88277$). The same witness gives $\mathrm{height}_P \ne 0$. Two steps of the withdrawn chain are independently invalid: the inequality $\|f\|_{\mathrm{Lip}} - \|[D_{\mathrm{CH}}, M_{K_{n_{\max}} * f}]\|_{\mathrm{op}} \le \|f\|_{\mathrm{Lip}} - \|K_{n_{\max}} * f\|_{\mathrm{Lip}}$ runs the wrong way, M being the *compressed* multiplier of §2.3 and compression contracting; and the good-kernel estimate bounds $\|K_{n_{\max}} * f - f\|_\infty$, not a difference of Lipschitz seminorms, which would require a second-order modulus.

Scope of the withdrawal. What falls is this lemma and the L5 route to Theorem 4.1. Theorem 1.7 is untouched: it uses only reach-type estimates and no height constituent, so the convergence statement — and with it the commutative-factor leg of the WH1 keystone — stands on that route. A height-controlled assembly for this pair is an open problem.

The assembly this proof used to close with is withdrawn with the height constituents. It read

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\le\; \max\big(\gamma_{n_{\max}}, \gamma_{n_{\max}}, \gamma_{n_{\max}}, 0\big) \;=\; C_3 \cdot \gamma_{n_{\max}},$$

whose third and fourth arguments are the refuted $\mathrm{height}_B \le \gamma_{n_{\max}}$ and $\mathrm{height}_P = 0$. **The conclusion (22) is therefore not established by this route.** What survives is Theorem 1.7, which bounds the same state-space GH distance by $\gamma_{n_{\max}}$ using reach-type estimates alone and no height constituent — so the *statement* of Theorem 4.1 stands, by that route, and the commutative-factor leg of the WH1 keystone with it (convergence of the almost-commutative structure is open, §5.2). □

Remark 3.13 (On the height term)**.** The height in (23) is a GeoVac-defined approximation constituent, loosely modelled on the height/reach of bridges in [16]; it is not a quantity defined in the spectral propinquity [15]. The naive bound $\|B_{n_{\max}}(f)\|_{\mathrm{op}} \le \|f\|_{L^\infty} \le 2\pi$ (from L4(b) plus the S^3 diameter in the dual-Coxeter metric) is not the relevant quantity here. Until 2026-09-03 the reduction of the height to $\gamma_{n_{\max}}$ [retracted 2026-09-04: l5-height-bound-achieved] via the Lipschitz-comparison form was presented as what gives the rate-controlled bound of Theorem 4.1; that reduction is refuted ((24)), and the bound comes from Theorem 1.7.

4 Main theorem

Theorem 4.1 (Main theorem; rate-controlled restatement)**.** ***[INTERNAL THEOREM]*** *Unconditionally (Theorem 1.7), the truncated Camporesi–Higuchi spectral triples $\mathcal{T}_{n_{\max}}$ converge to*

the round-S^3 Camporesi–Higuchi spectral triple $\mathcal{T}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance [8], under the translation-seminorm metrization of § 1.1:

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\le\; \gamma_{n_{\max}}, \qquad \gamma_{n_{\max}} = \frac{4\log n_{\max}}{\pi\, n_{\max}} + O\big(1/n_{\max}\big) \quad \textit{(dual-Coxeter metric of §2.1)}, \tag{25}$$

*equivalently $(2/\pi)\log n_{\max}/n_{\max} + O(1/n_{\max})$ on the unit S^3; the asymptotic constant is **[INTERNAL THEOREM + MEASURED]** (Lemma 3.4(d.ii)). **[OBSERVATION]** On the unit S^3 the asymptotic constant is the sphere-volume quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 2/\pi$ of unit-radius spheres; the dual-Coxeter constant $4/\pi$ is twice that quotient (numerically $\mathrm{Vol}(S^2)/\pi^2$). Neither is the Hopf fibration's base-to-total ratio, whose base carries half the radius (§3.2, volume-ratio note). **[PANEL-VERIFIED]** A uniform bound $\gamma_{n_{\max}} \le 6\log n_{\max}/n_{\max}$ holds for all $n_{\max} \ge 2$.*

Proof. Lemma 1.6 and Proposition 1.5 supply the operator-system substrate on the true window (Lemma 3.1 concerns the doubled computational model); Lemma 3.4 the central spectral Fejér kernel with closed-form $\gamma_{n_{\max}}$ and asymptotic rate; Lemma 3.7 the Lipschitz comparison constant $C_3 = 1$; Lemma 3.10 the Berezin reconstruction with the four properties a Latrémolière tunneling pair would require; and—in the *withdrawn* route only—Lemma 3.12 the assembly into the state-space GH bound.

Which route establishes this. Not the L5 assembly: its two height constituents are refuted ((24)), so (22) is *not* established by that route — as the withdrawal in Lemma 3.12 states, and as an earlier version of this proof failed to say, hedging instead only that the route "inherits the panel scope of Lemma 3.7". The bound comes from Theorem 1.7, which bounds the same state-space GH distance by $\gamma_{n_{\max}}$ using reach-type estimates alone and no height constituent, and which uses neither Lemma 3.7 nor Lemma 3.12. Combining that bound with the asymptotic and uniform rates of Lemma 3.4(d.ii)–(d.iii) gives (25). □

A companion proposition identifies the state-space GH limit explicitly.

Proposition 4.2 (Limit identification). ***[INTERNAL THEOREM]*** *Up to isometry, the state-space GH limit of the sequence $(\mathcal{T}_{n_{\max}})_{n_{\max}\ge 1}$ is*

$$\lim_{n_{\max}\to\infty} S(\mathcal{O}_{n_{\max}}) \;=\; \big(P(S^3),\, W_d\big),$$

where $P(S^3)$ is the space of Borel probability measures on S^3 and W_d is the Wasserstein–Kantorovich distance with respect to the dual-Coxeter geodesic distance d of §2.1; on the unit S^3, $W_{d_1} = W_d/2$.

Proof. Theorem 1.7 gives GH distance $\le \gamma_{n_{\max}} \to 0$ to $(S(C(S^3)), \mathrm{MK}_L)$, and a GH limit of compact metric spaces is unique up to isometry. States of $C(S^3)$ are the probability measures, and $L = \mathrm{Lip}_d$ (Lemma 1.3), so by Kantorovich–Rubinstein duality $\mathrm{MK}_L = W_d$. (The unit-metric Connes distance of D_{CH} gives $W_{d_1} = W_d/2$, a different, homothetic metric.) This is the same limit identification as in van Suijlekom [8] and Gaudillot-Estrada–van Suijlekom [7] for the scalar case. □

5 Discussion

5.1 Why SU(2) was harder than the torus

The flat-torus result of Leimbach–van Suijlekom [17] works on the abelian dual $\mathbb{Z}^d$ with ball truncations, a spectral Fejér kernel that is the square of the spherical Dirichlet kernel (not a product of circle Fejér kernels), and a Bożejko–Fendler-type transference, and proves convergence without an explicit rate. Three substantive complications arise on SU(2):

(i) Non-abelian Peter–Weyl. On SU(2) the Plancherel decomposition is not over $\mathbb{Z}^d$ but over half-integer weights $j \in \frac{1}{2}\mathbb{N}_{\geq 0}$ with isotypic blocks of dimension $2j+1$. The natural Plancherel weight $\sqrt{2j+1}$ is required to match the Fock-shell Peter–Weyl bijection $n = 2j+1$, and this is what produces the natural-coefficient kernel of Definition 3.3. (For central multipliers the cb-norm computation is elementary — block-scalar action — on any compact group; the genuine non-abelian difficulty is the Clebsch–Gordan structure of the multiplier symbol, not the cb-norm.)

(ii) No Kähler structure on S^3. The standard Berezin reconstruction of Hawkins [5] uses the holomorphic Toeplitz form on a Hardy space, which requires a compatible almost-complex structure. $S^3 = \mathrm{SU}(2)$ has no such structure (this is the standard fact that S^3 has trivial tangent bundle but no integrable almost-complex structure). The natural alternative is the central-spectral-Fejér-kernel form (19), which is the analog of van Suijlekom's [8] Fejér reconstruction on S^1, lifted to the non-abelian setting via the Peter–Weyl decomposition.

(iii) The SO(4) selection rule (reading withdrawn 2026-09-28). Earlier versions read the Lipschitz comparison as tight "because of" the multiplier sparsity of the SO(4) selection rule, with per-harmonic ratio $\sqrt{(N-1)/(N+1)}$, and contrasted it with the torus. That per-harmonic inequality is false (Lemma 3.7, remark), and the reading and the torus contrast are withdrawn: $C_3 = 1$ follows by compression on any closed spin manifold, with no sparsity input.

The $\log n$ factor in γ_n is the first-moment logarithm of the Fejér-type (squared-Dirichlet) kernel, which the circle already carries (Remark 3.5); it reflects the kernel choice, not non-abelianness. (Earlier versions called it the standard one-log non-abelian slowdown relative to a torus rate $O(1/n)$; Leimbach–van Suijlekom state no such rate, and the comparison is withdrawn, 2026-09-28.) **[OBSERVATION]** In the dual-Coxeter normalisation the asymptotic constant $4/\pi$ is twice the circle Fejér first-moment constant $2/\pi$; on the unit S^3 the two coincide, the factor 2 being the metric scale (Remark 3.5, 2026-09-03); we read the SU(2) value as the sphere-volume quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ at unit radius, i.e. $2\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ in the dual-Coxeter normalisation — both spheres of radius 1, so this is not the Hopf base-to-total ratio. (An earlier version of this sentence stated the constant as "identical on both sides"; withdrawn 2026-09-02.)

5.2 Marcolli–van Suijlekom gauge-network framework

The convergence theorem 4.1 is a statement about the continuum target S^3 — the commutative factor that almost-commutative spectral triple constructions hosted on S^3 take as input; convergence of the almost-commutative extension itself is open (below). The Marcolli–van Suijlekom gauge-network framework [18], with the Perez-Sanchez [28] continuum-limit correction, builds finite spectral triples from algebra-valued data on graphs; **[INTERNAL THEOREM]** the present convergence result concerns the Connes–van Suijlekom operator-system truncation of $\mathcal{T}_{S^3}$ on the true Camporesi–Higuchi window, which it identifies as metric-asymptotically the round-S^3 Camporesi–Higuchi geometry. That truncation is *not* the diagonal (gauge-network) representative $\mathbb{C}^{V_{\mathrm{Fock}}}$ of Paper 32, which is not the object that converges (Paper 32, scope clause of its GeoVac-triple theorem); the result therefore does not anchor the gauge-network construction itself. It supplies a continuum target — a geometry whose Dirac spectrum is independently known and whose spectral-action expansion is computable — against which a gauge-network construction on S^3 can be compared; whether any such construction converges to it is **[OPEN]**.

The natural next question — whether the almost-commutative extension $\mathcal{T}_{S^3} \otimes \mathcal{T}_F$ for a finite spectral triple $\mathcal{T}_F$ admits an analogous convergence statement — is open. **[OBSERVATION]** Inner-fluctuation analysis of the truncated AC extension at finite $n_{\max}$ has been carried out in supporting work by the present author on the chirality-doubled Fock model of the operator system (not the true window); the nonzero gauge-piece norms are measured, and the gauge piece is read as the expected $U(1)$ and SU(2) content under the Marcolli–van Suijlekom mechanism, but an operator-level identification with the gauge-network link variables is not established (`tests/test_almost_commutative.py`, Sprint H1; Paper 32 § Sprint H1), and the present paper

restricts attention to the geometric (single-factor) state-space GH statement.

5.3 Obstruction to a Riemannian–Hardy tensor product

A natural extension of Theorem 4.1 would aim at the tensor-product spectral triple $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\text{Bargmann}}$ obtained by tensoring the present Camporesi–Higuchi triple on S^3 with a Bargmann-segal-type spectral triple on S^5 in the holomorphic Hardy sector. **[OBSERVATION]** This extension is structurally obstructed at the framework level — the obstruction is argued from the categorical mismatch below, not proved as a theorem. The S^3 factor is Riemannian with KO-dimension 3 and a second-order Laplace–Beltrami structure; the S^5 factor is a Hardy-sector first-order complex Euler operator construction [23] which does not fit standard NCG Dirac axioms. The two factors live in different categories of NCG construction, and there is no published prescription for tensoring them. We have flagged this elsewhere as the Coulomb/HO asymmetry of Paper 24 (seven layers catalogued, six standing) resurfacing at the spectral-triple level. A proper Riemannian–Hardy propinquity theory is not in the literature and would require new framework development beyond the scope of the present paper.

5.4 Open questions

(i) Quantitative tightness. Theorem 4.1 gives the asymptotic constant $4/\pi$ in $\gamma_{n_{\max}} \sim (4/\pi)\log n_{\max}/n_{\max}$ but not the next-order constant. **[MEASURED]** Numerical doubling-estimator data suggests $n_{\max}\gamma_{n_{\max}} = (4/\pi)\log n_{\max} + b + O(\log n_{\max}/n_{\max})$ (a fit gives the next coefficient ≈ -0.64, consistent with the $\sim$1.85$\times$ rather than 2$\times$ residual shrink per doubling) with $b \approx 4.10$, whose closed form is undetermined at the level of the present analysis.

Update (2026-05-16): **[MEASURED]** the value b (rotation-angle normalisation) has been refined to 22 decimal digits (no frozen test in this repository computes b beyond six digits; the figure is a Paper 40 cross-reference) as $b = 4.1093214674898756204229\ldots$ (the original 80-digit string circulated in working memos prior to this paper was contaminated from digit 12 onward by float-precision panel points; corrected via re-extraction at ≥ 400 dps in a follow-on sprint, see [24] for details). PSLQ identification has been attempted at coefficient ceiling 10^8 against an extended L-derivative basis $(\zeta'(s), \beta'(s), \eta'(s), L'(2, \chi_q)$, Stieltjes constants, Glaisher–Kinkelin $\log A$) with NULL result. **[OPEN]** The closed form of b remains open, and the constant joins a small documented list of natural-but-unidentified constants in this programme, alongside the numerical coincidence formula studied in [25].

(ii) Lorentzian extension. The Camporesi–Higuchi spectral triple is Riemannian, with KO-dimension 3 (mod 8). On the Riemannian side, the finite-cutoff "period closure" $\sigma_{2\pi}(O) = O$ reported in the four-witness companion paper [21] (and [19] § VIII.F) is a trivial symbolic fact, not a modular-theoretic result: the generator used there is $K = 2J_z$ ($= J_{\text{polar}}$), which has integer spectrum, so $e^{2\pi i K} = 1$ and $\sigma_{2\pi}$ is the identity for *every* operator O; the four "witnesses" are the same K with the surface gravity relabelled, and nothing in the computation identifies a Bisognano–Wichmann modular flow of a wedge-restricted state (that test was not run). **[OPEN]** The genuinely Lorentzian extension at signature $(3,1)$ via Krein-space spectral triples is *open*, and is now known to be obstructed along its most natural route: compressing a Krein-self-adjoint product Dirac to the Krein-positive subspace annihilates the spatial Dirac, so the compressed Lipschitz seminorm vanishes identically on the natural operator system and no quantum-metric structure survives. The companion Lorentzian paper [22] proves this degeneracy theorem and states the repaired target (a Toeplitz temporal multiplier algebra in the Connes–van Suijlekom circle pattern, and a Krein-compatible Lipschitz seminorm that does not compress the Dirac).

(iii) Higher-rank compact non-abelian cases. The present proof transcribes mechanically to SU(n) for $n \geq 3$ at the algebraic level (Peter–Weyl, central characters, Plancherel weights) but, as originally posed, the Lipschitz comparison estimate was thought to require a

higher-rank version of the Avery 3-Y integral and the corresponding selection rule (the compression proof of Lemma 3.7, 2026-09-28, needs neither).

Update (2026-05-16): this open question has been resolved by Paper 40 [24], which generalises Theorem 4.1 to all compact connected semisimple Lie groups with bi-invariant metric (a circle factor carries the constant $2/\pi$ of Remark 3.5, so tori are excluded). The higher-rank analog of the Avery 3-Y selection rule is the *Dirac-triangle inequality* $|D(\pi) - D(\pi')| \le \sqrt{C(\sigma)}$ for every $\sigma \subset \pi \otimes \pi'^{*}$, which generalises Paper 38's $|n - n'| \le N - 1$ shell-difference rule (the literal "Casimir triangle inequality" $|C(\pi) - C(\pi')| \le C(\sigma)$ is false in general; the Dirac eigenvalue, not the Casimir, is the right ingredient). **[PANEL-VERIFIED]** The rate is *not* rank-dependent as anticipated above: the rate constant $4/\pi$ (dual-Coxeter normalisation) is reported *universal* across the class, by a Plancherel-weight × Vandermonde-Jacobian cancellation theorem (Paper 40 [24] §3.2) — rigorous at rank 1 (SU(2), the present paper), with the rank-≥ 2 leg corroborated on reduced panels and a fully rank-uniform analytical proof recorded as a named gap in Paper 40. **[INTERNAL THEOREM]** Asymptotic-tight $C_3(G) = 1$ at all ranks follows from the PRV-summand bound (Kumar 1988, Invent. Math. 93) together with a Brauer–Klimyk signed-sum closure for the interior summands, as proved in Paper 40 §3.3 (Lemma L3 there; the attribution to a 1990 Vinberg paper carried no locator and is withdrawn, 2026-09-03).

(iv) Beyond the commutative factor, and beyond the translation seminorm. [OPEN] Theorem 1.7 proves state-space convergence of the commutative factor on the true window under the translation seminorm. Convergence in the Dirac-commutator seminorm, and convergence of an almost-commutative (gauge-network, $\mathcal{A} \otimes M_n(\mathbb{C})$) structure built on it (§5.2), are not proved here.

A Detailed derivation of Lemma 3.4(d.ii)

[INTERNAL THEOREM + MEASURED] This appendix supplies the sum-rule / Euler–Maclaurin derivation (the "Stein–Weiss sharpening" in the corpus's naming, after the real-line kernel estimates of [29]) of the asymptotic $\lim_{n\to\infty} n\gamma_n/\log n = 4/\pi$ that appears as a sketch in the proof of Lemma 3.4. Step 3 isolates the two sub-coefficients (2 and -1) whose net is the logarithm's coefficient; each sub-coefficient and the net are pinned by independent doubling estimators (test-backed), which is what closes the constant. The argument is a standard asymptotic analysis of the closed-form sum-rule (16); we record it here in full because the referee may want to see the leading-order constant pinned without having to reconstruct it.

Setup. Recall the closed form of Lemma 3.4(d.i):

$$\gamma_n \;=\; \pi \;-\; \frac{4T_n}{\pi Z_n}, \qquad Z_n = \frac{n(n+1)}{2}, \qquad T_n = \sum_{\substack{1\le k_1,k_2\le n\\ k_1+k_2 \text{ odd}}} \sqrt{k_1 k_2}\Big[\frac{1}{(k_1-k_2)^2} - \frac{1}{(k_1+k_2)^2}\Big]. \tag{26}$$

The asymptotic claim of Lemma 3.4(d.ii) is equivalent to

$$\frac{T_n}{Z_n} \;=\; \frac{\pi^2}{4} \;-\; \frac{\log n}{n} \;+\; O\Big(\frac{1}{n}\Big), \tag{27}$$

since $\gamma_n = \pi - (4/\pi)(T_n/Z_n) = (4/\pi)(\log n/n) + O(1/n)$.

Step 1: Diagonal-dominant decomposition. Write $k_2 = k_1 + d$ with d odd, $d \in \{\pm 1, \pm 3, \dots\}$, $k_1 \in \{1, \dots, n\}$, $k_1 + d \in \{1, \dots, n\}$. By the $k_1 \leftrightarrow k_2$ symmetry of T_n, we may take $d > 0$ and double:

$$T_n \;=\; 2 \sum_{\substack{d>0\\ d \text{ odd}}} \sum_{a=1}^{n-d} \sqrt{a(a+d)}\Big[\frac{1}{d^2} - \frac{1}{(2a+d)^2}\Big].$$

Step 2: Leading-order $\pi^2/4$. The term $1/d^2$ dominates. Using $\sqrt{a(a+d)} = a\sqrt{1+d/a} = a + d/2 - d^2/(8a) + O(d^3/a^2)$ and $\sum_{a=1}^{n-d} 1 = n-d$,

$$\sum_{a=1}^{n-d} \sqrt{a(a+d)} \cdot \frac{1}{d^2} \;=\; \frac{1}{d^2}\Big[\frac{(n-d)(n-d+1)}{2} + \frac{d(n-d)}{2} + O(d\log n)\Big] \;=\; \frac{n^2}{2d^2} + O\Big(\frac{n}{d}\Big).$$

Summing over odd d from 1 to n and using $\sum_{d\,\mathrm{odd}} 1/d^2 = \pi^2/8$ (the Catalan-Euler-style identity $\sum_{d\geq 1\,\mathrm{odd}} 1/d^2 = (1-2^{-2})\zeta(2) = (3/4)(\pi^2/6) = \pi^2/8$),

$$T_n \;=\; 2\cdot\frac{n^2}{2}\cdot\frac{\pi^2}{8} + \text{lower order} \;=\; \frac{\pi^2 n^2}{8} + \text{lower order}.$$

Dividing by $Z_n = n(n+1)/2 \sim n^2/2$ gives the leading $T_n/Z_n \to \pi^2/4$, confirming $\gamma_n \to 0$.

Step 3: Sub-leading $\log n$ (two sub-coefficients, $2-1=1$). Write $\mathrm{defect}_n := \pi^2/4 - T_n/Z_n$, so that (27) is the statement $\mathrm{defect}_n = \log n/n + O(1/n)$, i.e. that the $\log n/n$ coefficient of the defect is exactly 1 (and $\gamma_n = (4/\pi)\,\mathrm{defect}_n$ identically by (26)). Reorganise the Step-1 double sum by a, with d odd and $d \leq n-a$, and expand $\sqrt{a(a+d)} = a + d/2 + r(a,d)$. Two pieces carry a logarithm:

(A) *Triangle truncation of the d-sum.* The $a\cdot(1/d^2)$ piece is $2\sum_{a=1}^{n-1} a\,S_2(n-a)$ with $S_2(D) = \sum_{d\,\mathrm{odd}\leq D} d^{-2} = \pi^2/8 - 1/(2D) + O(D^{-2})$ (Euler–Maclaurin on the odd-$\zeta(2)$ tail). The tail contributes $-\sum_{a=1}^{n-1} a/(n-a) = -nH_{n-1} + O(n)$, so this piece alone gives defect log-coefficient 2.

(B) *The $d/2$ correction.* The $(d/2)\cdot(1/d^2)$ piece is $\sum_{a=1}^{n-1} S_1(n-a)$ with $S_1(D) = \sum_{d\,\mathrm{odd}\leq D} d^{-1} = \frac{1}{2}\log D + O(1)$, which is $\frac{1}{2}n\log n + O(n)$; it *adds* to T_n/Z_n and so reduces the defect with log-coefficient 1.

The remaining pieces — the $-1/(2a+d)^2$ bracket and the residual $r(a,d)$ — carry no $\log n$ at leading order: the net coefficient on the exact T_n is $2-1=1$. **[MEASURED]** All three statements are pinned independently by doubling (Richardson) estimators on the genuinely computed partial sums, none of which assumes the answer: the piece-(A) estimator converges to 2, the piece-(B) estimator to 1, and the estimator on the exact defect_n to 1 (`tests/test_trunk_qa_fejer_4_over_pi.py`: `test_subcoeff_A_triangle_truncation_is_two`, `test_subcoeff_B_sqrt_correction_is_one`, `test_net_defect_coefficient_converges_to_one`). Hence

$$\frac{T_n}{Z_n} \;=\; \frac{\pi^2}{4} \;-\; \frac{\log n}{n} \;+\; O\Big(\frac{1}{n}\Big),$$

which is (27). (An earlier version of this step attributed the logarithm to the $-1/(2a+d)^2$ bracket and to a boundary term in $\sqrt{a(a+d)}$, each "$-\pi^2 n\log n/16$"; that bookkeeping did not compose to the displayed result and is replaced by the test-backed decomposition above, 2026-09-02.)

Step 4: Conversion to γ_n. Substituting into (26),

$$\gamma_n \;=\; \pi \;-\; \frac{4}{\pi}\Big(\frac{\pi^2}{4} - \frac{\log n}{n} + O(1/n)\Big) \;=\; \frac{4\log n}{\pi n} \;+\; O\Big(\frac{1}{n}\Big),$$

which is the asymptotic of Lemma 3.4(d.ii).

Numerical verification. **[MEASURED]** The closed form $n\gamma_n = (4/\pi)\log n + b + O(\log n/n)$ with $b \approx 4.109$ has been verified to high precision against the sum-rule (26) at $n \in \{2, \ldots, 1000\}$, with the doubling estimator $a_n := (2n\gamma_{2n} - n\gamma_n)/\log 2 \to 4/\pi$ reaching 1.2785 at $n = 800$ and 1.2761 at $n = 1600$ (three digits of $4/\pi$; residual shrinking $\sim 1.85\times$ per doubling) (`tests/test_trunk_qa_fejer_test_gamma_doubling_converges_to_4_over_pi`, `test_su2_doubling_estimator_at_n800` and the slow-marked `test_su2_doubling_estimator_at_n1600`; the sum-rule itself is checked against Gaussian quadrature of the kernel moment in `tests/test_central_fejer_su2.py`).

Relation to the circle Fejér estimate. The constant $4/\pi$ is *twice* the constant of the classical estimate $\frac{1}{2\pi}\int_{\mathbb{T}^1} F_n(\theta)\,|\theta|\,d\theta \sim (2/\pi)\log n/n$ for the probability-normalised Fejér kernel on the circle (Zygmund [30] Vol. I, Ch. III; Remark 3.5, where the circle constant is computed in closed form and measured). The $\sin^2(\chi/2)$ factor in the SU(2) conjugacy-class measure and the $\sqrt{2j+1}$ Plancherel weighting of Definition 3.3 together change the kernel and the constant while preserving the leading $\log n/n$ behaviour. **[OBSERVATION]** On the circle the constant is $2\mathrm{Vol}(S^0)/\mathrm{Vol}(S^1) = 2/\pi$; on SU(2) it is $\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2)) = 2/\pi$ on the unit S^3 and $2\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2)) = \mathrm{Vol}(S^2)/\pi^2 = 4/\pi$ in the dual-Coxeter normalisation; no common "2Vol(base)/Vol(group)" form is claimed (2026-09-03), and no derivation of one constant from the other is given. (An earlier version of this paragraph stated the circle constant as $4/\pi$ and the SU(2) constant as "preserved across this transition"; withdrawn 2026-09-02.)

Acknowledgments

The author thanks Alain Connes, Walter van Suijlekom, François Latrémolière, Matilde Marcolli, Eva-Maria Hekkelman, Edward McDonald, Malte Leimbach, and Carlos Perez-Sanchez for the foundational frameworks on which this paper builds; remaining errors are the author's own. This work was developed in an AI-augmented agentic workflow with Anthropic's Claude: implementation of the internal proof-assistant verification machinery, drafting of the internal proof memos used as supporting documentation, and bibliographic collation were performed collaboratively. All theorem statements, design choices, and editorial decisions are the author's responsibility. The author thanks the open-source NCG community (notably the SageMath, sympy, and mpmath teams) for the computational infrastructure used in the supporting numerical verifications.

References

[1] Z.-Y. Wen and J. Avery, *Some properties of hyperspherical harmonics*, J. Math. Phys. **26** (1985), 396–403.

[2] R. Camporesi and A. Higuchi, *On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces*, J. Geom. Phys. **20** (1996), 1–18.

[3] A. Connes, *Noncommutative geometry and reality*, J. Math. Phys. **36** (1995), 6194–6231.

[4] A. Connes and W. D. van Suijlekom, *Spectral truncations in noncommutative geometry and operator systems*, Comm. Math. Phys. **383** (2021), 2021–2067; arXiv:2004.14115.

[5] E. Hawkins, *Quantization of equivariant vector bundles*, Comm. Math. Phys. **202** (1999), 517–546; and *Geometric quantization of vector bundles and the correspondence with deformation quantization*, Comm. Math. Phys. **215** (2000), 409–432.

[6] E.-M. Hekkelman, *Truncated geometry on the circle*, Lett. Math. Phys. **112** (2022), 20; arXiv:2111.13865.

[7] Y. Gaudillot-Estrada and W. D. van Suijlekom, *Convergence of spectral truncations for compact metric groups*, Int. Math. Res. Not. IMRN **2025**, no. 13 (2025), rnaf197; arXiv:2310.14733.

[8] W. D. van Suijlekom, *Gromov–Hausdorff convergence of state spaces for spectral truncations*, J. Geom. Phys. **162** (2021), 104075; arXiv:2005.08544.

[9] M. A. Rieffel, *Matrix algebras converge to the sphere for quantum Gromov–Hausdorff distance*, Mem. Amer. Math. Soc. **168** (2004), no. 796, 67–91; arXiv:math/0108005.

[10] C. Farsi and F. Latrémolière, *Collapse in noncommutative geometry and spectral continuity*, arXiv:2404.00240, 2024.

[11] C. Farsi and F. Latrémolière, *Continuity for the spectral propinquity of the Dirac operators associated with an analytic path of Riemannian metrics*, arXiv:2504.11715, 2025.

[12] E. Hekkelman and E. McDonald, *A noncommutative integral on spectrally truncated spectral triples, and a link with quantum ergodicity*, J. Funct. Anal., to appear (2025), arXiv:2412.00628.

[13] R. Toyota, *Quantum Gromov–Hausdorff convergence of spectral truncations for groups with polynomial growth*, arXiv:2309.13469, 2023.

[14] F. Latrémolière, *The dual Gromov–Hausdorff propinquity*, J. Math. Pures Appl. **103** (2015), 303–351.

[15] F. Latrémolière, *The Gromov–Hausdorff propinquity for metric spectral triples*, Adv. Math. **404** (2022), Paper No. 108393, 56pp; preprint arXiv:1811.10843, 2018.

[16] F. Latrémolière, *The quantum Gromov–Hausdorff propinquity*, Trans. Amer. Math. Soc. **368** (2016), 365–411.

[17] M. Leimbach and W. D. van Suijlekom, *Gromov–Hausdorff convergence of spectral truncations for tori*, Adv. Math. **439** (2024), Paper No. 109496.

[18] M. Marcolli and W. D. van Suijlekom, *Gauge networks in noncommutative geometry*, J. Geom. Phys. **75** (2014), 71–91; arXiv:1301.3480.

[19] J. Loutey, *The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage*, GeoVac internal preprint (2026), DOI via Zenodo.

[20] V. G. Kac, *Infinite-Dimensional Lie Algebras*, 3rd ed. Cambridge University Press, 1990.

[21] J. Loutey, *Tomita–Takesaki modular structure on truncated $SU(2)$ spectral triples: the four-witness 2π period at finite cutoff*, GeoVac internal preprint (2026), DOI via Zenodo.

[22] J. Loutey, *Lorentzian propinquity on truncated* $\mathrm{SU}(2) \times U(1)_T$ *Krein spectral triples: a degeneracy theorem*, GeoVac internal preprint (2026), DOI via Zenodo.

[23] J. Loutey, *The Bargmann–Segal Lattice: π-Free Discretization of the Harmonic Oscillator on S^5*, GeoVac internal preprint (2026), DOI via Zenodo.

[24] J. Loutey, *State-space Gromov–Hausdorff convergence of spectral truncations of compact semisimple Lie groups with bi-invariant metric*, GeoVac internal preprint (2026), DOI via Zenodo. Generalizes the present paper from SU(2) to compact connected *semisimple* Lie groups (a circle factor carries the circle constant); rate constant $4/\pi$ reported universal

across that class (dual-Coxeter normalisation; rigorous at rank 1, rank-uniform proof a named gap) via a Plancherel-weight × Vandermonde-Jacobian cancellation; asymptotic-tight $C_3 = 1$ at all ranks via the PRV-summand bound (Kumar 1988) with a Brauer–Klimyk interior closure.

[25] J. Loutey, *The Fine Structure Constant as a Spectral Coincidence on the Hopf Fibration*, GeoVac internal preprint (2026), DOI via Zenodo.

[26] J. Loutey, *The Dimensionless Vacuum: Recovering the Schrödinger Equation on a Scale-Invariant Label Lattice*, GeoVac internal preprint (2026), DOI via Zenodo.

[27] J. Loutey, *Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds. Master Mellin engine §III.7*, GeoVac internal preprint, papers/group3_foundations (2026), DOI via Zenodo.

[28] C. I. Perez-Sanchez, *Comment on 'Gauge networks in noncommutative geometry'*, arXiv:2508.17338, 2025.

[29] E. M. Stein and G. Weiss, *Introduction to Fourier Analysis on Euclidean Spaces*, Princeton Math. Ser. **32**, Princeton Univ. Press, 1971.

[30] A. Zygmund, *Trigonometric Series, Vol. I and II*, 3rd ed., Cambridge Univ. Press, 2002.